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Single-cell RNA-seq reveals trans-sialidase-like superfamily gene expression heterogeneity in *Trypanosoma cruzi* populations

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eLife Assessment

This study provides **important** insights into how *Trypanosoma cruzi* populations diversify surface protein expression, showing through single-cell RNA sequencing that trans-sialidase-like genes are expressed heterogeneously across individual parasites, a pattern with clear implications for immune evasion. The evidence is **convincing**, supported by robust single-cell transcriptomic analyses, consistent quantitative measures of expression heterogeneity, and integration with genomic organization that together argue against purely stochastic expression.

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Abstract

Trypanosoma cruzi, the causative agent of Chagas disease, presents a major public health challenge in Central and South America, affecting approximately 8 million people and placing millions more at risk. The *T. cruzi* life cycle includes transitions between epimastigote, metacyclic trypomastigote, amastigote, and blood trypomastigote stages, each marked by distinct morphological and molecular adaptations to different hosts and environments. Unlike other trypanosomatids such as *Trypanosoma brucei*, *T. cruzi* does not employ a monoallelic model of antigenic variation; instead, it relies on a diverse repertoire of cell-surface associated proteins encoded by large multigene families, which are essential for infectivity and immune evasion.

This study analyzes cell-specific transcriptomes using single-cell RNA sequencing of amastigote and trypomastigote cells to characterize stage-specific surface protein expression during mammalian infection. Through clustering and identification of cell-specific markers, we assigned cells to distinct parasite developmental forms. Analysis of individual cells revealed that surface protein-coding genes, especially members of the trans-sialidase like superfamily (TcS), are expressed with greater heterogeneity than single-copy genes. Moreover, no recurrent combinations of TcS genes were observed between individual cells in the population. Remarkably, a small subset of TcS mRNAs, encoded by genes preferentially located in the core genomic compartment, are frequently detected across the cell population, whereas the vast majority of TcS

mRNAs show low detection frequencies and are mainly encoded in the disruptive compartment. Our findings thus reveal transcriptomic heterogeneity within trypomastigote populations where each cell displays unique TcS expression profiles. Focusing on the diversity of surface protein expression, this research aims to deepen our understanding of *T. cruzi* cellular biology and infection strategies.

Introduction

The protozoan parasite *Trypanosoma cruzi* is the etiological agent of Chagas disease, a highly prevalent infectious disease in Central and South America that affects approximately 8 million people, with several million more at risk of infection (1).

The parasite life cycle involves both an invertebrate vector (triatomine bug) and a vertebrate host. The change in environmental conditions triggers differentiation processes in the parasite developing across four main stages. The epimastigote form replicates within the insect and differentiates into metacyclic trypomastigotes in the insect's rectal tract. This latter form does not replicate, specializes in host infection, and can invade various cell types. Once inside the cell, the parasite differentiates into an amastigote, the replicative form within the mammalian host. After several rounds of division, amastigotes differentiate into bloodstream trypomastigotes, which, after cell lysis, can either infect new cells or be ingested by the vector (2). Parasite developmental stages exhibit significant morphological and molecular differences, associated with distinct gene expression profiles. In trypanosomatids, protein-coding genes are organized into large polycistronic RNA polymerase II transcription units, with limited evidence for tight, gene-specific transcriptional control (3). As a result, gene expression is largely regulated at the post-transcriptional level, through mechanisms controlling mRNA steady-state abundance, translation, and related processes (3). More recently, accumulating evidence indicates that genome organization and chromatin state also shape transcriptional outputs (4–8). These studies have identified two distinct chromatin compartments in *T. cruzi*: a core compartment, in which genes tend to show more uniform expression consistent with permissive transcription and predominant post-transcriptional regulation (9), and a disruptive compartment, enriched in multigene families, whose genes exhibit lower average transcription and stronger chromatin-associated regulation (4).

T. cruzi infection relies on a heterogeneous set of membrane proteins, encoded mainly by large multigene families (10). Among these are trans-sialidases and trans-sialidase-like proteins (TcS), mucins, MASPs, GP63, DGFs, and RHS proteins, most of which are involved in infection, tropism, and immune evasion (11–16).

The trans-sialidase-like superfamily is involved in processes underlying host-parasite interactions (12). Members that retain enzymatic activity catalyze the transfer of sialyl groups from host glycoconjugates to galactopyranosyl units on the parasite's surface, an essential activity, as *T. cruzi* is unable to synthesize *de novo* sialic acid (17). This is the largest superfamily, with over 1,400 members, and is subdivided into eight groups based on amino acid sequence (12). Few members have been functionally characterized, most expressed primarily in mammalian stages (18). The TcS group I include proteins that retain trans-sialidase activity, though members from all groups are involved in host-parasite interactions (19).

In recent years, single-cell RNA sequencing (scRNA-seq) has been employed in protozoa, with reports including *Plasmodium falciparum*, *Trypanosoma brucei*, and *Leishmania major* (20–29). These studies revealed key aspects of the infection process undetectable with conventional methods, highlighting the relevance of this approach for understanding individual variation in gene expression in single-celled organisms (30). In *T. brucei*, antigenic variation driven by variant surface glycoproteins (VSGs) has been studied at single-cell resolution to understand the mechanisms that enable subpopulations of this parasite to evade the immune system, as a bet hedging strategy, ensuring parasite survival (25). In this parasite, scRNA-seq revealed that pre-metacyclic cells express multiple VSG transcripts simultaneously, in contrast to metacyclic forms, which display a protein coat composed of a single VSG type.

While population-level heterogeneity in surface protein expression has been suggested as critical for *T. cruzi* infection, immune evasion, and persistence (31,32), this has not been studied at the intra-population level, and the underlying mechanisms remain poorly understood. Here, we present a scRNA-seq analysis of *T. cruzi*, to understand the heterogeneity in surface protein expression within trypomastigote populations.

Results and Discussion

Identification of cell populations

To assess the expression of cell-surface protein-coding genes in *Trypanosoma cruzi*, we conducted a 10X Chromium Single Cell 3' assay from a mixed population of amastigotes and cell-derived trypomastigotes, aiming at sequencing the transcriptome of 5000 cells. After low-quality cell filtering and gene expression quantification (see Materials and Methods), we obtained 3192 single-cell transcriptomes with 14321 total genes detected, with a mean of 1088 genes and 2461 UMIs detected per cell. These results were comparable to other scRNA-seq studies done in *Trypanosoma brucei* and more recently addressed in *T. cruzi* (currently reported as a preprint) using 10X Chromium technology (26,33,34). Cell populations (Figure 1a) were defined by identifying cluster-specific gene markers (Figure 1b and Supplementary File 1). Two cell clusters were assigned to trypomastigotes and amastigotes: cluster 0 (2201 cells) and cluster 1 (824 cells), respectively. Markers gene expression was consistent with previous bulk RNA-seq data from Dm28c (4) (Figures 1b, 1c and 1d, Supplementary Table 1). In turn, we hypothesize that cluster 2, that comprised only 167 cells, reflects amastigote-trypomastigote transitioning parasites, as its gene markers are differentially expressed in bulk RNA-seq data, but some are upregulated in amastigotes and others in trypomastigotes (Supplementary Table 1).

Expression pattern of surface protein-coding genes

We analyzed differences in gene expression between single-copy genes and multigene families, between and within the identified cell populations. As previously reported (4,18), multi gene family expression is increased in trypomastigotes (Figure 2a), consistent with the involvement of these genes in stage-specific functions. As control, we examined the expression of single-copy genes, for which this pattern was not observed as these genes are primarily associated with basic cellular functions and show more similar average expression levels between the two cell types (Supplementary Figure 1a)(33).

Within trypomastigotes, heterogenous expression of surface protein-coding gene families was observed, with high variation (compared to single-copy genes) in the number of cells in which each surface protein-coding gene was detected, as well as the total expression level in each cell (Figure 2b and 2c). Even though expression heterogeneity is also observed for single-copy genes (Supplementary Figure 1a and Figure 1b), probably due to the sampling biases that cause gene dropout in 10X Genomics technology, we investigated whether this phenomenon was more pronounced in surface multigene families. Therefore, we analyzed differences among single-copy and multigene family genes (together or grouped by multigene family) in trypomastigotes, in terms of the number of cells expressing each of the individual genes of each group (Figure 2c, Supplementary Figure 1c and Supplementary Table 2). In addition, we assessed expression inequality using Lorenz curves (Figure 2d, Supplementary Figure 1d and Supplementary Table 2) to evaluate how unevenly gene expression is distributed within each gene group. Ribosomal protein-coding genes were included as a control group.

Interestingly, compared to single-copy genes, especially ribosomal genes, multigene family genes showed a greater dispersion regarding the number of cells in which each gene was detected (Figure 2c, Supplementary Figure 1c and Supplementary Table 2), as well as a more pronounced deviation from the diagonal in the Lorenz curves (Figure 2d, Supplementary Figure 1d), which represents perfect equality (reflected by different Gini indexes, see Supplementary Table 2). Both observations indicate a higher expression heterogeneity for surface protein expression in the trypomastigote population. When multigene families were analyzed separately,

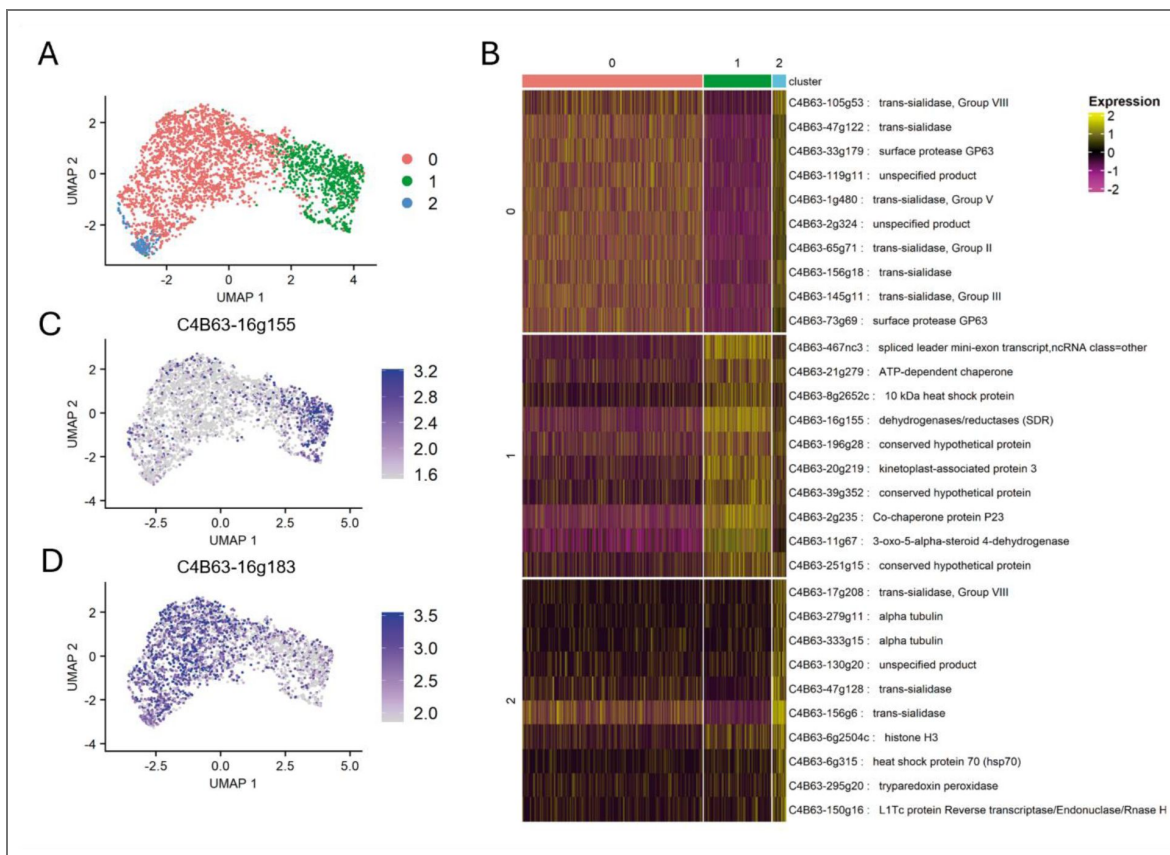


Figure 1. Identification of amastigote and trypomastigote cell populations.

(a) UMAP colored by detected clusters based on gene expression profiles. (b) Heatmap of the top 10 gene markers upregulated in each of the 3 cell populations identified ($\log_2FC > 1$ and adjusted p-value < 0.05). (c) Expression of a cluster 0 marker gene (C4B63-16g155) on the UMAP, and (d) Expression of a cluster 1 marker gene (C4B63-16g183) on the UMAP.

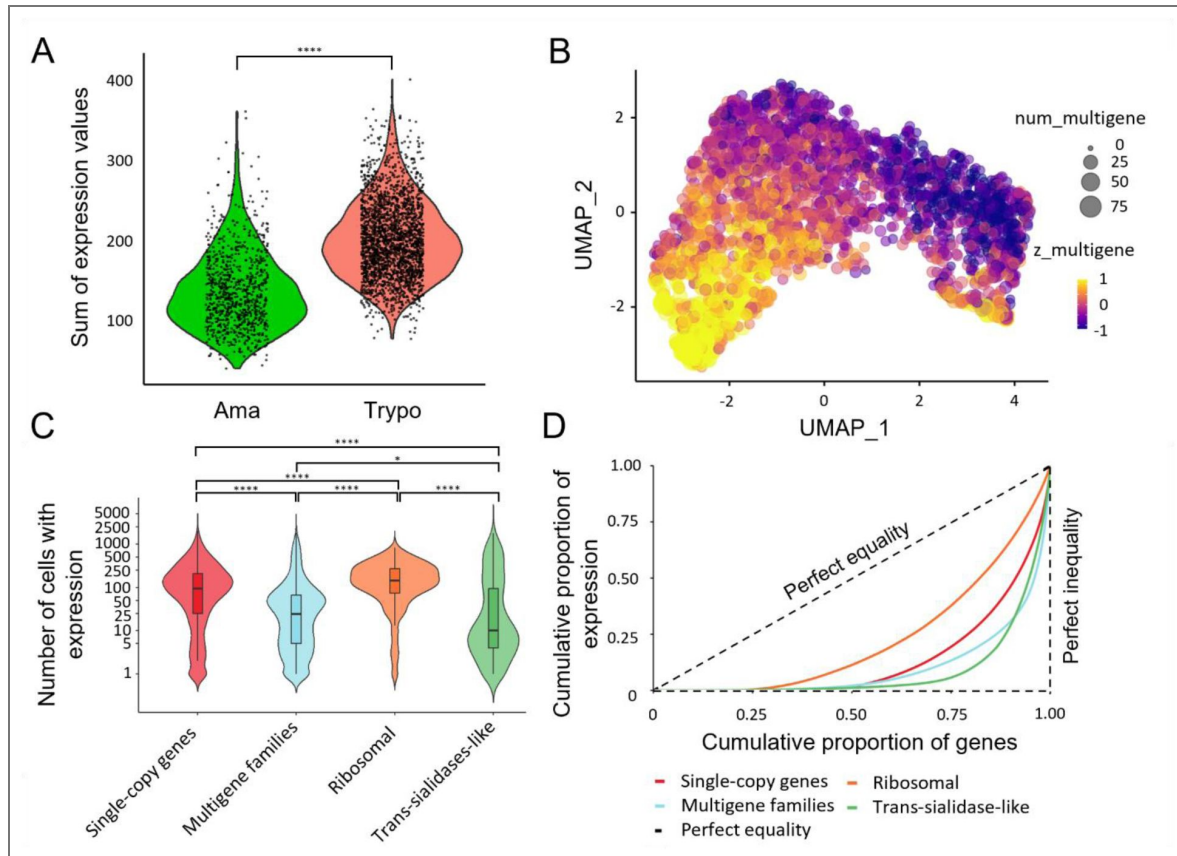


Figure 2. Overview of expression patterns across amastigote and trypomastigote cells.

(a) Summatory of expression levels values from all multigene family genes for each cell from amastigote (Cluster 1) and trypomastigote (Cluster 0) cell populations (**** $p < 0.0001$, $\text{mean}_{\text{Ama}} = 137.2$, $\text{SD}_{\text{Ama}} = 48.7$, $\text{mean}_{\text{Trypo}} = 201.1$, $\text{SD}_{\text{Trypo}} = 48.6$, $\text{FC}_{\text{Trypo/Ama}} = 1.5$). (b) UMAP visualization of the expression patterns of multigene family genes; num_multigene indicates the number of multigene family genes detected per cell (genes with >0 UMI counts). Z_multigene reflects the relative expression level of multigene family genes per cell, calculated as the z-score-standardized sum of their UMI counts, such that positive values reflect above-average multigene family expression and negative values reflect below-average levels. (c) Violin plots showing the number of cells expressing a specific gene belonging to each group of genes: subsampled single-copy and multigene families, ribosomal genes and trans-sialidases. To avoid biases against size differences between single-copy and multigene family genes we generated a subsampled single-copy genes list, randomly selecting an equal number of genes as those from the multigene family's gene set. The expression distribution of the subsampled single-copy genes is similar to the distribution of the entire dataset (* $p < 0.05$, **** $p < 0.0001$. See Supplementary Table 2). (d) Lorenz curves showing the cumulative proportion of gene expression relative to the cumulative proportion of genes for subsampled single-copy, multigene family genes, ribosomal protein coding genes and trans-sialidase-like genes. Genes were ordered by total expression, and the dashed line indicates perfect equality. Curves that deviate further from the diagonal reflect greater inequality, meaning that fewer genes account for most of the expression within each category. Statistically significant differences between groups for c) and d) are shown in Supplementary Table 2.

several showed significant differences when compared to single-copy genes. Heterogeneity of the MASP family has already been suggested by studying clonal populations supporting our single cell results (31). Also, TcS genes exhibited a pronounced expression heterogeneity (Supplementary Figures 1c and 1d). TcS constitute a well-established and biologically central gene family in *T. cruzi*, playing key roles in host-parasite interactions (11,12,17). In this context, we next focused on a more detailed characterization of TcS expression heterogeneity in trypomastigotes.

Expression pattern of the TcS superfamily

When we re-clustered the trypomastigote population based solely on TcS gene expression, we identified two sub-populations: trypomastigote cluster 0 ("Trypo_0" composed of 1186 cells), which over-expressed these genes compared to trypomastigote cluster 1 ("Trypo_1" composed of 1015 cells) (Figure 3a).

The two trypomastigote sub-populations segregated by TcS expression show only slight differences in the expression of other gene categories, such as ribosomal protein-coding genes (FC = 1.03), transporters (FC = 1.09), polymerase-related genes (FC = 1.15), and phosphatases (FC = 1.10) (Figure 3b and Supplementary Figure 2), foldchange values correspond to Trypo_0/Trypo_1 expression ratios). Although there is a tendency for surface protein genes to be more expressed in cluster Trypo_0 (FC = 1.38), trans-sialidases displayed the highest fold change between the two trypomastigote subpopulations (FC = 1.53) (Figure 3c and Supplementary Figure 2). It is tempting to speculate that this may reflect different infectivity amongst trypomastigote subpopulations, consistent with reports of "broad" and "slender" forms (35). Recently, while this manuscript was being prepared, a similar approach by Laidlaw et al. described the phenomenon by clustering trypomastigote cells based on the expression of all genes (34). When applying this strategy to our data, their observation was reproduced.

When all cells are considered, most TcS genes are expressed, but in each individual cell only approximately 40 TcS genes were detected. Interestingly, we observed that both trypomastigote subpopulations contain a subgroup of TcS genes that are detected in a large portion of cells (>40%) (Figure 4a, and Supplementary Table 3), indicating high-level expression at the population level. This subset comprises 31 TcS genes belonging to subfamilies II-VI, and VIII (Supplementary Table 3). Consistent with our findings, Laidlaw et al. (34) reported a similar phenomenon, detecting on average approximately 40 TcS genes per cell, with a subset that are frequently detected across the trypomastigote population, further supporting the reproducibility of scRNA-seq results across studies. Gene dropouts in scRNA-seq experiments can generate apparently stochastic detection patterns for multigene family members, particularly affecting low-abundance transcripts, and thus represent an important potential confounder in the analysis of multigene families. To evaluate whether this technical effect could account for the observed TcS detection patterns, we examined the relationship between detection frequency and expression level. If random sampling were the primary driver, genes with similar average expression levels would be expected to exhibit comparable detection frequencies. However, we observed that many TcS genes with similar average expression are detected in markedly different proportions of cells (Figure 4b), arguing against a purely stochastic dropout model. Notably, among the 50 TcS genes with the highest average expression across cells, 62% are detected in fewer than 5% of cells (Figure 4b).

At the single-cell level, we found that detected TcS genes contribute relatively evenly to the total family expression within each cell, as indicated by the low average Gini index of TcS expression levels across each cell (Figures 4c and 4d). This suggests that, within each cell, total TcS expression is shared across the detected TcS genes rather than being dominated by a few very highly expressed transcripts in that cell. Consistent with this interpretation, most TcS genes detected in a high fraction of cells in our scRNA-seq dataset also rank among the most highly expressed TcS genes in an independent bulk RNA-seq study (Supplementary Figure 3a). This highlights a key limitation of bulk RNA-seq, as it may wrongly indicate that a set of few genes are highly expressed in each cell, while in fact, TcS expression is evenly distributed among all detected

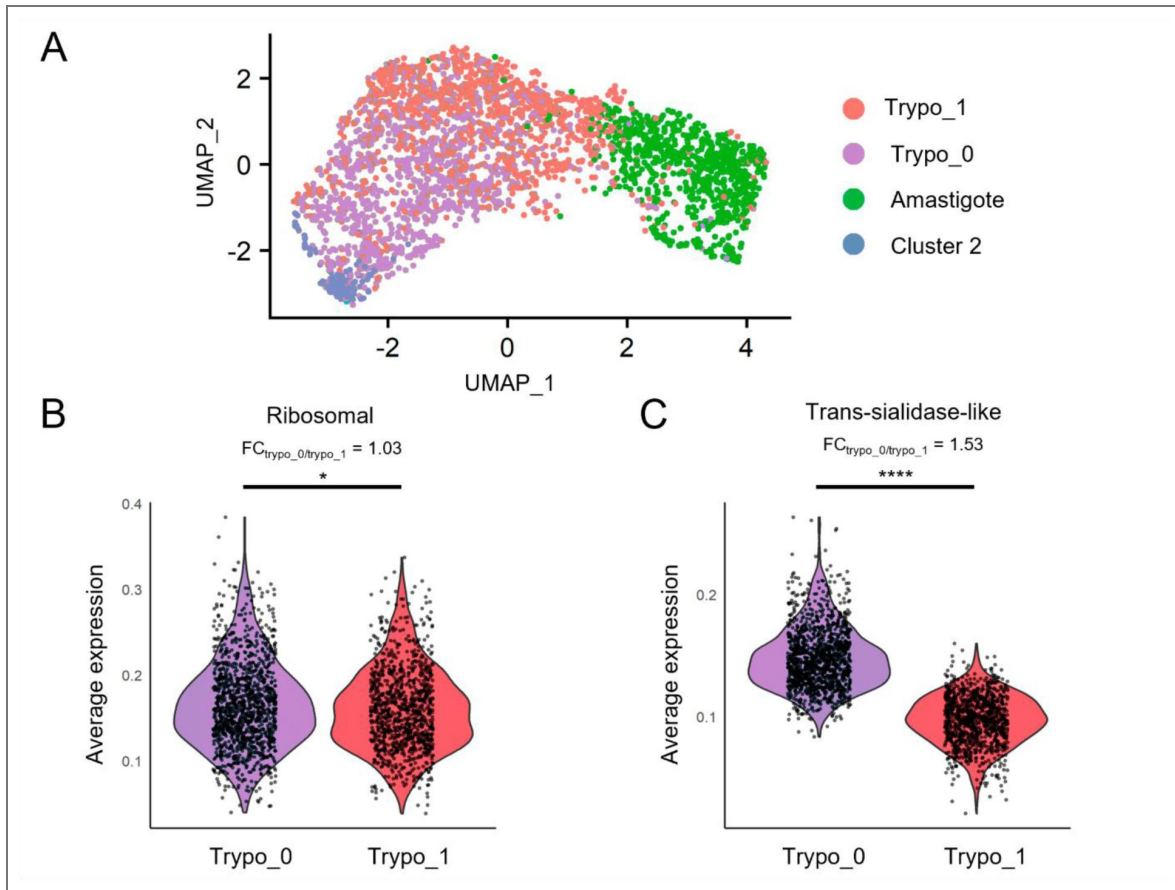


Figure 3. Trypomastigote sub-populations identified based on trans-sialidase expression profiles.

(a) UMAP visualization colored by detected clusters based on gene expression profiles, with trypomastigote subpopulations identified. (b) Violin plot displaying average expression levels of ribosomal protein-coding genes across sub-populations. (c) Violin plot showing combined trans-sialidase expression levels for each sub-population. * $p < 0.05$, **** $p < 0.0001$.

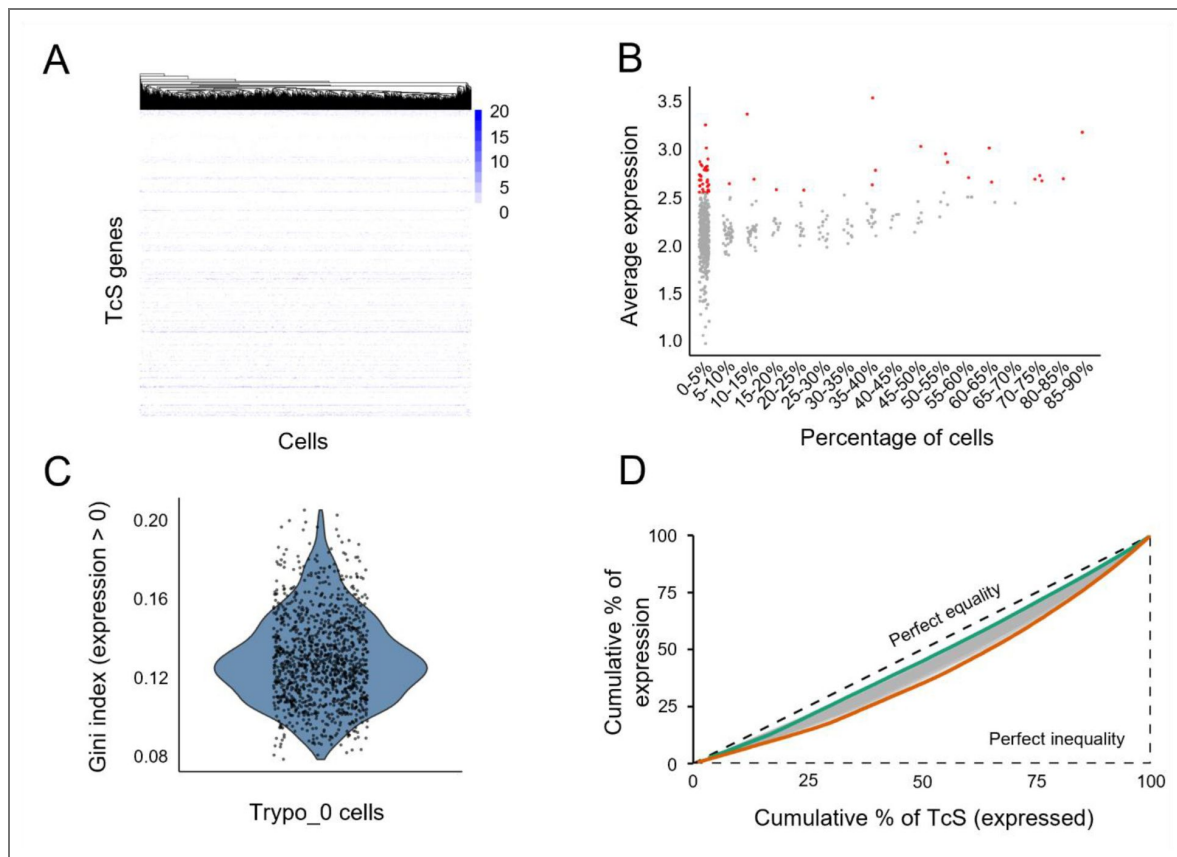


Figure 4. Overview of TcS gene expression patterns in Trypo_0 cells.

(a) Heatmap displaying the expression of TcS genes in each cell that together account for 75% of total TcS gene expression within cluster Trypo_0. Cells are clustered by TcS expression profiles, with colors representing each gene's percentage contribution to the cell's total TcS expression. (b) Average expression of TcS genes grouped by the percentage of cells expressing each gene. In red are highlighted the top 50 TcS with highest average expression. (c) Gini index distribution for trypomastigotes cluster 0 (Trypo_0) cells considering only TcS detected in each cell. (d) Lorenz curves showing, for each cell in cluster Trypo_0, the cumulative proportion of total TcS expression as a function of the cumulative proportion of detected TcS genes. Genes were ordered by total expression, and the dashed line indicates perfect equality (i.e., all detected TcS genes contribute equally to the total TcS expression of a given cell). Green and orange curves correspond to cells with higher and lower expression equality, respectively.

family members, regardless of the number of cells expressing each gene. Taken together with the results in Figure 4b, this observation further supports the conclusion that the observed expression pattern is unlikely to arise from a purely stochastic detection of TcS mRNAs.

While our data indicates transcriptional heterogeneity, it is important to note that future studies will be required to determine the extent to which this diversity translates to protein expression on the parasite surface. Consistent with our observations, recent proteomic analyses (36) report pronounced variability in the expression of surface proteins, including TcS, suggesting that the heterogeneous transcriptional patterns observed here likely reflect biologically relevant differences.

When analyzing each trypomastigote subpopulation, no coordinated expression among specific TcS members was observed, as no subclusters of cells were identified based on TcS detection profiles. Even when clustering was restricted to genes detected in more than 40% of cells, no clear subclusters of cells were identified (Supplementary Figure 3b).

As discussed in the Introduction, although post-transcriptional control remains a central mechanism of gene expression regulation in *T. cruzi*, increasing evidence in recent years has revealed an important contribution of epigenetic mechanisms that modulate gene expression at the transcriptional level. (4–8). Specifically, the loci for TcS genes and other gene families of surface proteins are mostly grouped in specific genomic compartments (9) that are regulated epigenetically by the activation or silencing of chromatin folding domains (4). The TcS that are detected in a high percentage of cells are mostly dispersed throughout the genome (Supplementary Table 3). This suggests that their preferential expression is likely not due to colocalization in one or a few ubiquitous activated chromatin-folding domains. Nevertheless, mapping the genomic locations of TcS genes detected in a high proportion of cells revealed that most are flanked by core genes (Figure 5). The core compartment is enriched in conserved, single-copy genes that typically show more constitutive expression (9) and, as observed in this study, lower cell-to-cell variability. In contrast, TcS genes that are detected less frequently are preferentially located in the disruptive compartment (Figure 5), which is enriched in lineage-specific multigene families and associated with more variable, stage-specific, and potentially stochastic expression under tighter epigenetic control (4,9,18,36). Together, these findings suggest that the higher cellular prevalence of certain TcS transcripts is unlikely to be driven by colocalization within a small number of ubiquitously active chromatin domains but may instead reflect distinct regulatory regimes between the core and disruptive compartments. Future studies integrating single-cell chromatin profiling with scRNA-seq will be required to directly test this model.

Final remarks

The expression of surface protein-coding genes varied across parasite developmental stages. In particular, genes belonging to the TcS superfamily, which play key roles in host-parasite interactions, exhibited marked heterogeneity in expression among trypomastigotes. Notably, while most TcS genes were detected only in a small fraction of cells, a limited subset was frequently detected across the population. This pattern indicates that TcS expression is not uniform, suggesting the existence of distinct regulatory regimes within the family. Consistent with previous single-cell and bulk RNA-seq studies (34,36), the lack of coordinated expression among TcS members points to a complex regulatory framework that may enable functional diversification among *T. cruzi* subpopulations. Importantly, our analyses indicate that the frequent detection of this TcS subset may be partially explained by their preferential localization within the core genomic compartment, which is associated with more permissive transcriptional environments.

Taken together, our results demonstrate the sensitivity of scRNA-seq for resolving parasite life stages and their associated transcriptional programs, while also pointing to complex regulation of surface protein expression, particularly within the TcS family. Although these results should be considered with caution, as technical limitations inherent to scRNA-seq may influence the observed expression patterns, our interpretation is consistent with all observations presented

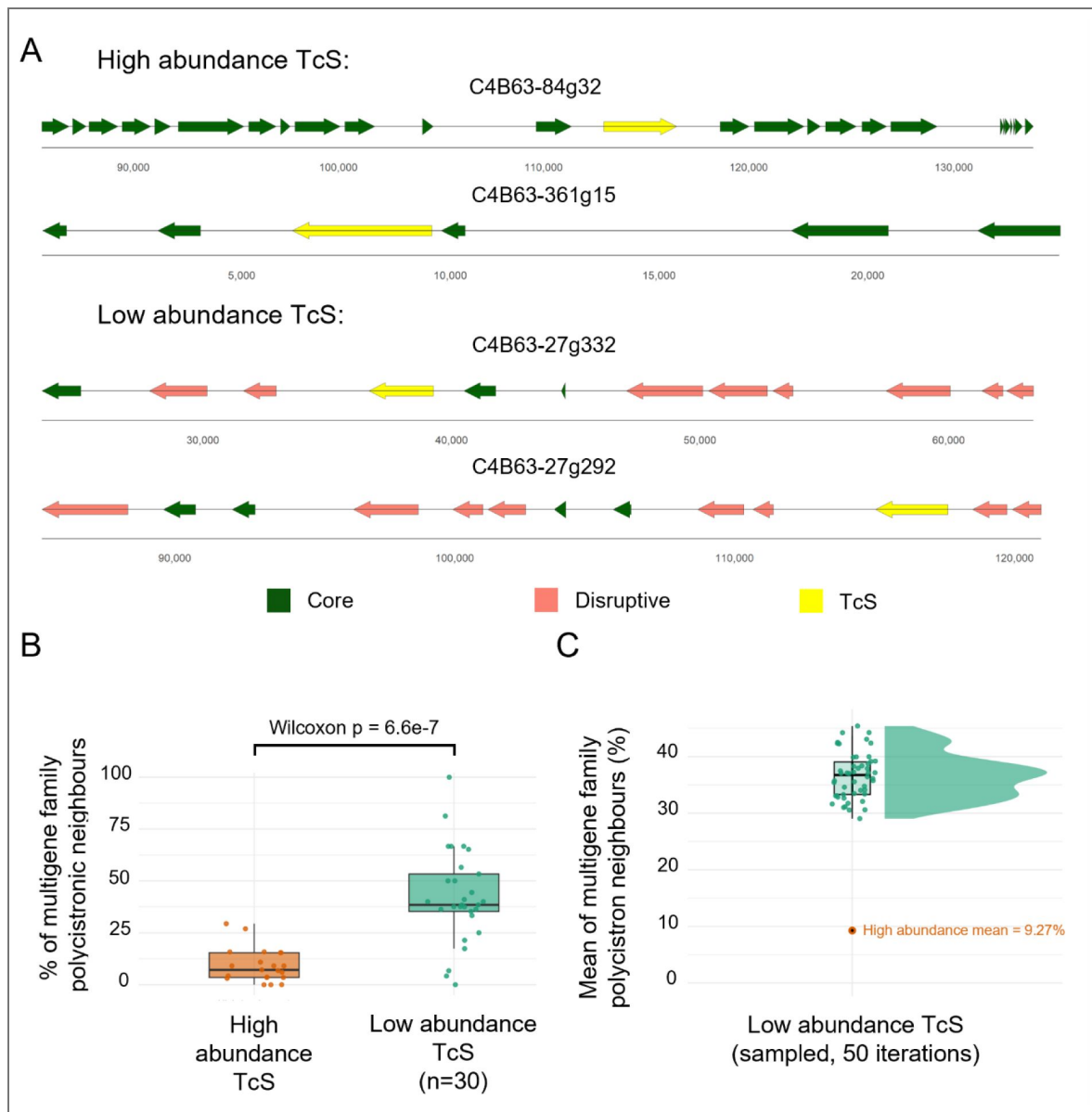


Figure 5. Genomic context and neighborhood composition of frequently detected versus lowly detected TcS loci.

(a) Representative genomic loci of frequently detected (top, high abundance) and lowly detected (bottom, low abundance) TcS genes. Genes are shown as arrows, colored according to genomic compartment: core (dark green), disruptive (salmon), and TcS genes under analysis (yellow). Chromosomal coordinates are indicated below each locus. (b) Comparison of the percentage of multigene-family neighbours within polycistronic transcription units containing frequently detected and lowly detected TcS genes. Lowly detected TcS genes were subsampled to $n = 30$. Wilcoxon rank-sum test: $p = 6.6 \times 10^{-7}$. (c) Mean percentage of multigene-family neighbours in polycistrons calculated from 50 random subsets ($n = 30$) of lowly detected TcS genes (mean = 37.76%). The orange dot indicates the corresponding mean for frequently detected TcS genes (9.27%).

here and aligns with emerging evidence from independent studies. Collectively, this supports a working hypothesis in which heterogeneous TcS expression may contribute to immune evasion and pathogenicity through a bet-hedging strategy.

Materials and Methods

Trypanosoma cruzi and mammalian cell culture

Epimastigote forms of *Trypanosoma cruzi* strain Dm28c were derived from axenic cultures cultivated in Brain-Heart Infusion medium (BHI, Oxoid) supplemented with 10% heat-inactivated fetal bovine serum (FBS, Capricorn), penicillin (100 units/mL) and streptomycin (100 µg/mL) as described (37). Cultures were diluted 1/10 with fresh BHI medium every 3 days and maintained at 28°C.

Myoblast rat cell line H9c2 (ATCC CRL-1446) was maintained in hgDMEM medium (Gibco) supplemented with 10% heat-inactivated fetal bovine serum, penicillin (100 units/mL) and streptomycin (100 µg/mL) at 37°C in a humidified 5% CO₂ incubator. Confluent cells were washed with 1X phosphate-buffered saline (1X PBS), incubated for 5 min with trypsin-EDTA (Gibco), diluted with culture medium and re-plated for maintenance.

Mycoplasma contamination in cell lines was regularly monitored using MycoAlert® Mycoplasma Detection Kit (Lonza), following the manufacturer's protocol.

Isolation and purification of cellular trypomastigotes and intracellular amastigotes

Late stationary phase epimastigotes of Dm28c strain were used to infect H9c2 cells for a primary infection. Six days post-infection, cellular trypomastigotes were obtained from the supernatant and were used to infect 50% confluent H9c2 cells at a 10:1 rate. Twenty-four hours post-infection the cell culture was washed twice with PBS 1X to remove any remaining extracellular parasites and maintained with fresh hgDMEM at 37°C in a humidified 5% CO₂ incubator.

For amastigote purification, infected H9c2 cells incubated for 48 hours post infection were washed with 1X PBS and incubated with trypsin-EDTA for 5 minutes at 37°C. The trypsinization was stopped by adding an equal volume of hgDMEM with 10% FBS. The cell suspension was repeatedly passed through a 27-gauge needle attached to a 30-mL syringe until complete cell disruption was confirmed under the microscope. The supernatant, containing free amastigotes, was collected and centrifuged at 500xg for 10 minutes at 4°C to remove large host-cell debris. The resulting supernatant was then centrifuged at 4,000xg for 10 minutes at 4°C, and the amastigote-containing pellet was washed twice in chilled 1X DPBS (Dulbecco's Phosphate-Buffered Saline, No Calcium, No Magnesium) and resuspended in 1X DPBS at 200 µL per 1×10⁶ cells, ready for the fixation step.

Cellular trypomastigotes derived from infected H9c2 cells and present in the cell supernatant fraction were collected and centrifuged at 500xg for 10 minutes at 4°C to remove large host-cell debris. The washing and resuspension steps in DPBS were performed as previously described for amastigotes.

scRNA-seq library preparation and sequencing

Cell fixation was performed using the Methanol Fixation Protocol for Single-Cell RNA Sequencing (38), after resuspension in DPBS as described above, as recommended by 10X Genomics technical support. Briefly, chilled 100% Methanol (for HPLC, ≥99.9%, Millipore) was added drop by drop (1×10⁶ cells in 800 µL) and incubated at -20°C for 30 min. For rehydration, fixed cells were first equilibrated at 4°C and then centrifuged at 4,000xg for 5 min at 4°C, the supernatant was discarded and Wash-Resuspension Buffer (3X SSC in Nuclease-free Water, 0.04% UltraPure Bovine Serum Albumin, 1mM DTT, and 0.2 U/ml RNase Inhibitor) was added to the pellet. Cell debris and large clumps were eliminated by passing the sample through a 40 µm Flowmi Cell Strainer.

10X Genomics library preparation was performed at the service provided by the Instituto de Biología y Medicina Experimental (IBYME, Argentina). The library was sequenced by a service provider (Macrogen, Korea) in a HiSeq2500 equipment (two lanes), generating approximately 880 million reads of 91 bp. Raw data is available in the SRA (<https://www.ncbi.nlm.nih.gov/sra/>) under BioProject PRJNA1200704.

Transcript quantification

T. cruzi 2018 Dm28c genome (9) (release 62, TriTrypDB (39)) and *T. cruzi* Dm28c maxi circle kDNA sequence (40) were combined and used as reference. To improve the proportion of reads assigned to genes, 11,362 3'UTR regions of the coding sequences (CDS) were annotated using peaks2UTR (41). Gene expression quantification was performed by pseudoalignment using kallisto bustools (42), with the options `--filter` to remove potential noise from environmental RNA and `--em` to apply the Expectation-Maximization (EM) algorithm. Default values were used for the rest of the parameters. This algorithm outperforms other mapping software in handling multimapping reads, enabling more accurate quantification of multigene families (42). This resulted in 321 million reads mapped to the transcriptome.

scRNA-seq data processing and analysis

Count matrices from two technical replicates obtained by sequencing the same library were merged: the common barcodes across both datasets were retained, and the count matrices were combined to generate a unified dataset. Subsequently, the following metrics were calculated: nUMI, nGene, and mitoRatio, as well as the log10 of genes per UMI (log10GenesperUMI). A filtering criterion was applied, retaining cells with nUMI > 1200, nGene > 100, log10GenesperUMI > 0.8, and mitoRatio < 0.1. Ribosomal rRNA genes were excluded from subsequent analyses. Data normalization and scaling were performed using the Seurat R version 5 package (43), employing `NormalizeData` and `ScaleData` functions.

After quality filtering, 3,192 single-cell transcriptomes were retained with an average of 8004 reads mapped per cell. In total, 14,321 genes were detected across all cells (93.5% of the 15,319 annotated protein-coding genes in the *T. cruzi* 2018 Dm28c genome). Per cell, we observed a mean of 1,088 detected genes and 2,461 UMIs, which corresponds to ~7.1% of the annotated protein-coding gene.

`FindNeighbors` function was used to construct a k-nearest neighbours graph of cells using 10 principal components (PCs) and Louvain algorithm was employed for cell clustering using `FindClusters` function. Additionally, doublets were identified and removed using the `DoubletFinder` R package (44); only 41 doublets were detected, all belonging to cluster 2.

To define marker genes, the `FindAllMarkers` function from Seurat package was used, selecting those with an adjusted p-value < 0.05 and a log2 fold change (log2FC) > 1. To validate these markers, bulk RNA-seq data of Dm28c was incorporated (NCBI BioProject ID PRJNA850400 [24]). Transcripts were quantified using Kallisto (45) (with `-b 100` option to perform 100 bootstraps), followed by differential expression analysis conducted with Sleuth (46). Genes with an adjusted p-value < 0.05 and $|\log_2FC| > 0.25$ were filtered for further analysis. Finally, the gene IDs of the Seurat-defined markers were cross-referenced with the IDs of the differentially expressed genes obtained from the bulk RNA-seq analysis to corroborate stage-specific gene expression. For 2D visualization of cell clusters and gene markers expression profiles across cells, UMAP projection was employed (47).

Gene IDs corresponding to multigene families (TcS, MASP, Mucins, GP63, RHS and DGF) were obtained by text searches using the current genome annotation on the “description” field. Throughout the manuscript, these genes are referred to as either surface protein genes or multigene family genes. Single and low copy number genes were defined as those that did not belong to the latter gene list. For simplicity “single-copy” will be used to refer to these genes throughout the manuscript. To avoid biases arising from differences in list size between single-

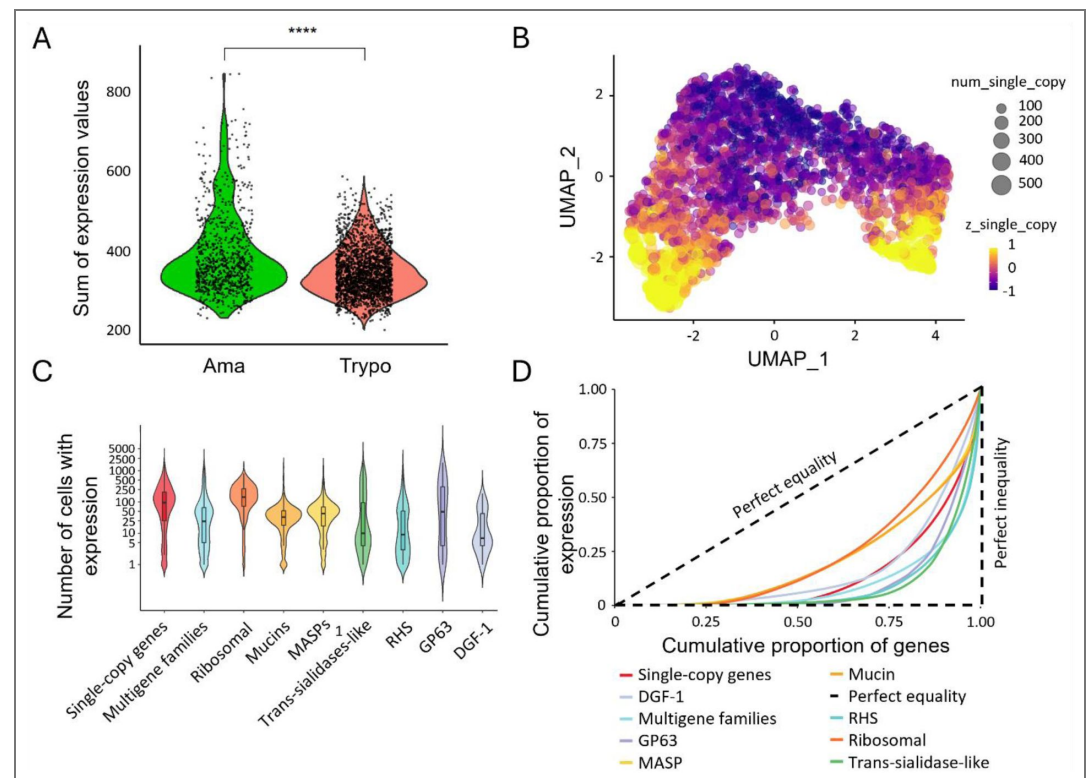
copy and multigene family gene sets, we generated a subsampled single-copy gene list by randomly selecting the same number of genes as in the multigene family set. The expression distribution of this subsampled single-copy gene set is similar to that of the full dataset.

Data processing was conducted using R version 4.2.0. Statistical analyses were performed using the Wilcoxon rank-sum test and p-values < 0.05 were considered statistically significant.

To study expression inequality, we plotted Lorenz curves (48) and applied the Gini index, a metric originally developed in the field of economics (49). Both metrics serve as a measure of expression heterogeneity, and we employed it in two distinct contexts: first, to evaluate the degree of inequality in the distribution of a gene's expression levels across individual cells (Figure 2d, Supplementary Figure 1d and Supplementary Table 2); and second, to assess the extent to which a given cell expresses individual genes at varying rates (Figure 4c).

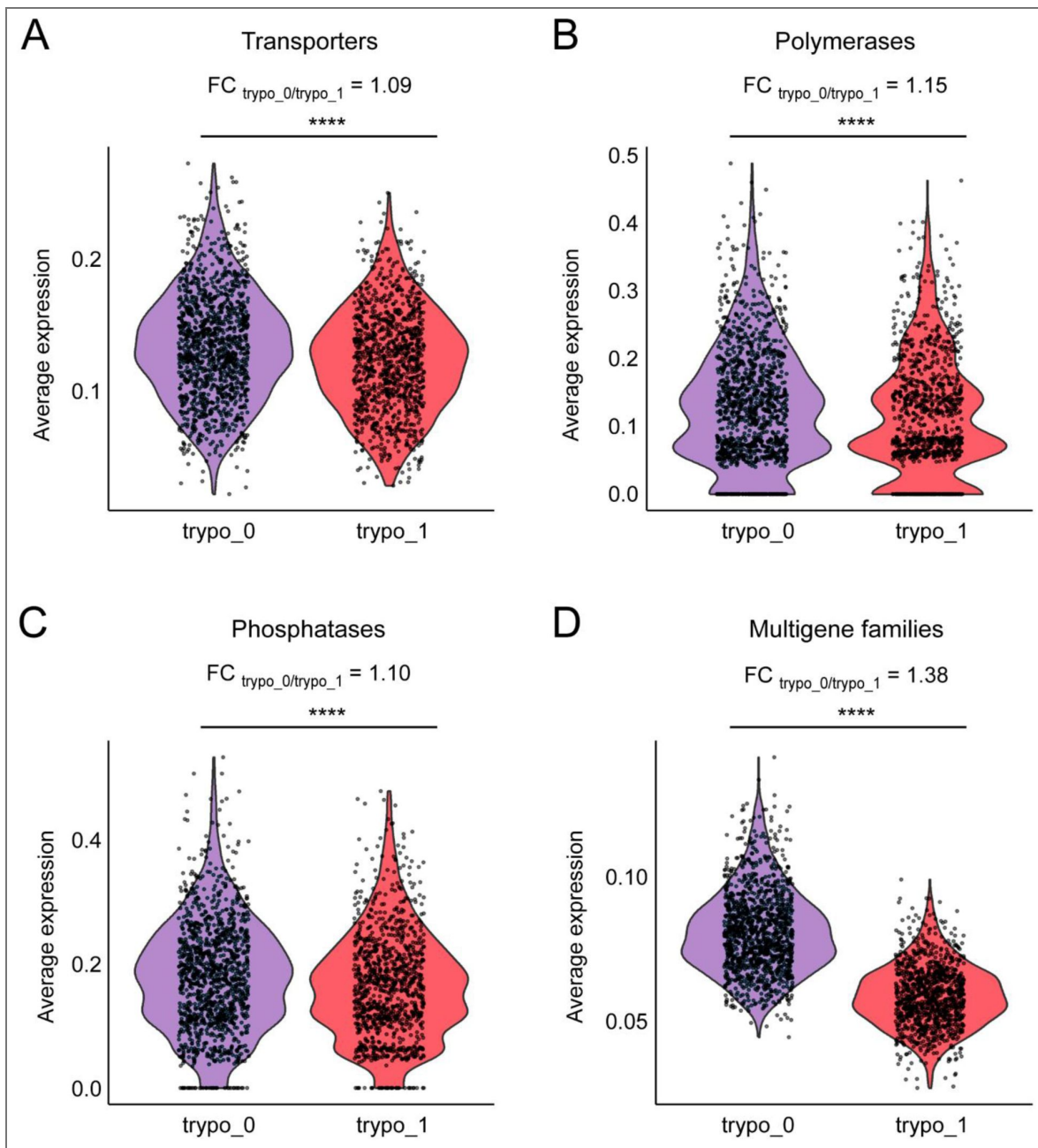
TcS genomic localization analysis was performed by defining directional gene clusters from the GFF annotation: protein-coding genes were ordered by coordinate per contig and consecutive genes on the same strand were grouped into a DGC, with a strand-switch (change from "+" to "-" or vice versa) marking DGC boundaries. For each gene, we calculated the percentage of its polycistronic neighbours (genes within the same DGC) that belong to the multigene family gene set previously defined and compared high-abundance TcS vs low-abundance TcS (sampling n = 30). For statistical comparisons, we assessed the robustness of the analysis by sampling 50 iterations of n = 30 low-abundance TcS and distributions were compared to the high-abundance set using two-sided Wilcoxon rank-sum tests with Benjamini-Hochberg correction.

Supplementary Figures



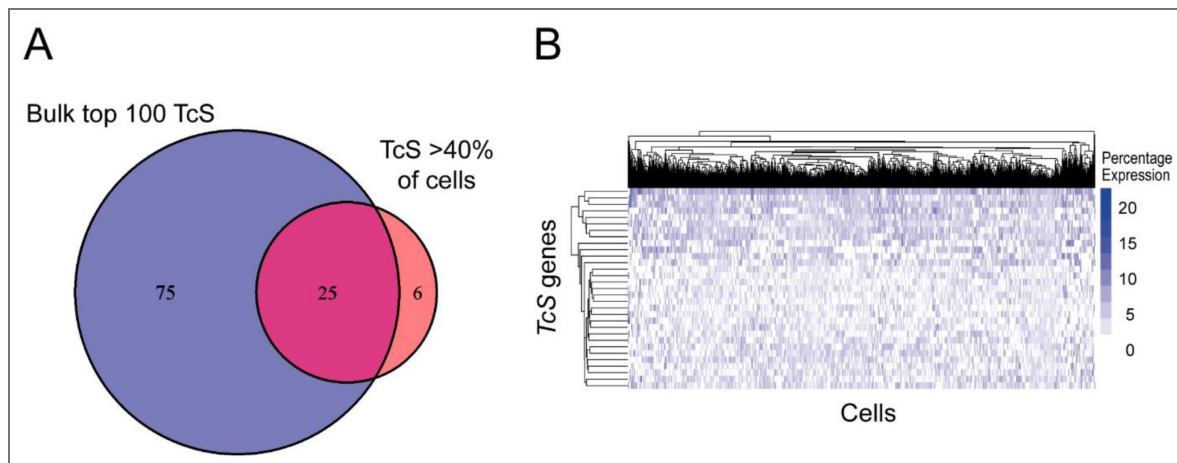
Supplementary Figure 1. (a) Summatory of expression levels values from subsampled single-copy genes for each cell from amastigote (Cluster 1) and trypomastigote (Cluster 0) cell populations (**** p < 0.0001, meanAma = 394.9, SDAma = 101.7, meanTrypo = 353.1, SDTrypo = 62.1, FCAma/Trypo = 1.1). (b) UMAP projection for 2D visualization of core gene expression among cells; num_multigene indicates the number of multigene family genes detected in each cell, whereas z_multigene indicates the expression levels calculated by summing the UMI counts of all multigene family genes in each cell and then standardizing this value using a z-score transformation, such that positive values reflect above-average multigene family expression and negative values reflect below-

average levels. (c) Violin plots showing the number of cells expressing a specific gene belonging to each group of genes: subsampled single-copy and multigene families, ribosomal genes and different multigene families. (d) Lorenz curves showing the cumulative proportion of gene expression relative to the cumulative proportion of genes for subsampled single-copy, multigene family genes (together or grouped by multigene family) and ribosomal protein coding genes. Genes were ordered by total expression, and the dashed line indicates perfect equality. Curves that deviate further from the diagonal reflect greater inequality, meaning that fewer genes account for most of the expression within each category. Statistically significant differences between groups for c) and d) are shown in Supplementary Table 2.



Supplementary Figure 2. Trypomastigote sub-clusters identified based on trans-sialidase expression profiles.

Violin plots displaying average expression levels across sub-clusters and associated fold changes ($FC_{\text{trypano}_0/\text{trypano}_1}$) of (a) transporters coding genes, (b) DNA and RNA polymerase-associated protein coding genes, (c) phosphatases coding genes and (d) multigene family genes. **** $p < 0.0001$.



Supplementary Figure 3.

(a) Venn diagram showing the overlap between the top 100 most expressed TcS from bulk RNA-seq data and TcS expressed in more than 40% of cells from cluster tryo_0, and (b) Heatmap displaying the expression of TcS genes in each cell that together account for 75% of total TcS gene expression and are expressed in more than 40% of cells within cluster tryo_0. Cells are clustered by TcS expression profiles, with colors representing each gene's percentage contribution to the cell's expression.

Data availability

Raw sequence data is available in the SRA (<https://www.ncbi.nlm.nih.gov/sra/>) under BioProject PRJNA1200704.

Additional information

Author Contributions

LI: Data Curation, Formal Analysis, Investigation, Methodology, Visualization, Writing-Original Draft Preparation, Writing-Review & Editing; LB: Methodology, Investigation, Writing-Review & Editing; VAC: Methodology, Investigation, Writing-Review & Editing; JG: Methodology, Formal Analysis, Writing-Review & Editing; GR: Methodology, Writing-Review & Editing; VMH: Methodology, Writing-Review & Editing; MAD: Methodology, Resources, Writing-Review & Editing; JSS: Methodology, Resources, Writing-Review & Editing; JDG: Conceptualization, Methodology, Visualization, Writing-Original Draft Preparation, Writing-Review & Editing; PS: Conceptualization, Funding Acquisition, Methodology, Formal Analysis, Project Administration, Resources, Supervision, Validation, Writing-Original Draft Preparation, Writing-Review & Editing

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Additional files

[Supplementary File 1](#)

[Supplementary Table 1](#)

[Supplementary Table 2](#)

[Supplementary Table 3](#)

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Peer reviews

Reviewer #1 (Public review):

Summary:

The authors aimed to assess the variability in expression of surface protein multigene families between amastigote and trypomastigote *Trypanosoma cruzi*, as well as between individuals within each population. The analysis presented shows higher expression of multigene family transcripts in trypomastigotes compared to amastigotes and that there is variation in which copies are expressed between individual parasites. Notably, they find no clear subpopulations expressing previously characterised trans-sialidase groups and that no patterns of coexpressed TcS genes were evident within individual cells or subpopulations. They also note that TcS encoded in the core genome are more often expressed, compared to TcS genes encoded in other genome compartments.

Strengths:

Additionally, the authors successfully process methanol fixed parasites with the 10x Genomics platform. This approach is valuable for other studies where using live parasites for these methods is logistically challenging.

In this second submission the authors show the kallisto mapping approach used is as robust as possible, and that this approach outperforms STAR mapping.

Weaknesses:

The authors describe a single experiment, which lacks repeats, controls or complementation with other approaches and the investigation is limited to the trans-sialidase transcripts.

Comments on revised version:

Thank you to the authors for taking the time to thoroughly address the peer review. The main concerns have now been addressed, and the manuscript edited to make points of confusion clearer.

<https://doi.org/10.7554/eLife.105822.2.sa3>

Reviewer #2 (Public review):

Summary:

This manuscript presents a valuable single-cell RNA-seq study on *Trypanosoma cruzi*, an important human parasite. It investigates the expression heterogeneity of surface proteins, particularly those from the trans-sialidase-like (TcS) superfamily, within amastigote and trypomastigote populations. The findings suggest a previously underappreciated level of diversity in TcS expression, which could have implications for understanding parasite-host interactions and immune evasion strategies. The use of single cell approaches to delve into population heterogeneity is strong. However, the study does have some limitations that need to be addressed.

The focus on single-cell transcriptional heterogeneity in surface proteins, especially the TcS family, in *T. cruzi* is novel. Given the important role of these proteins in parasite biology and host interaction, the findings have potential significance.

Strengths:

The key finding of heterogeneous TcS expression in trypomastigotes is well-supported. The analysis comparing multigene families, single-copy genes, and ribosomal proteins highlights the unusual nature of the variation in surface protein coding genes.

Weaknesses:

While the manuscript identifies TcS heterogeneity, the functional implications of the different expression profiles remain speculative. The authors state it may reflect differences in infectivity, but no direct experimental evidence supports this.

The manuscript lacks any functional validation of the single-cell findings. For instance, do the trypomastigote subpopulations identified based on TcS expression exhibit differences in infectivity, host cell tropism, or immune evasion? Such experiments would greatly strengthen the study.

The authors identify a subpopulation of TcS genes that are highly expressed in many cells. However, it is unclear if these correspond to previously characterized TcS members with specific functions.

The authors hypothesize that observed heterogeneity may relate to chromatin regulation. However, the study does not directly address these mechanisms. There are interesting connections to be made with what they identify as colocalization of genes within chromatin folding domains, but the authors do not fully explore this. It would be insightful to address these mechanisms in future work. [...]

Comments on revisions:

The novel version of the manuscript has improved and satisfied this reviewer.

<https://doi.org/10.7554/eLife.105822.2.sa2>

Reviewer #3 (Public review):

The study aimed to address a fundamental question in *T. cruzi* and Chagas disease biology - how much variation is there in gene expression between individual parasites? This is particularly important with respect to the surface protein-encoding genes, which are mainly from massive repetitive gene families with 100s to 1000s of variant sequences in the genome. There is very little direct evidence for how expression of these genes is controlled. The authors conducted a single cell RNAseq experiment of in vitro cultured parasites with a mixture of amastigotes and trypomastigotes. Most of the analysis focused on the heterogeneity of gene expression patterns amongst trypomastigotes. They show that heterogeneity was very high for all gene classes, but surface-protein encoding genes were the most variable. Interestingly, in the case of the trans-sialidase genes, many sequence variants were detected in fewer than 5% of parasites while a subset of 31 others was detected in >40% of parasites, hinting at compartmentalised expression control within the gene family. The biology of the parasite (e.g. extensive post-transcriptional regulation) and potential technical caveats (e.g. high dropout rates across the genome) make it difficult to infer connections to actual protein expression on the parasite surface, but the results are a significant advance for the field.

(1) Limit of detection and gene dropouts.

An average of ~1100 genes are detected per parasite which indicates a dropout rate of over 90%. It appears that RNA for the "average" single copy 'core' gene is only detected in around 3% of the parasites sampled (Figure 2c: ~100 / 3192). While comparable with some other trypanosome scRNAseq studies, this remains a caveat to the interpretation that high cell-to-cell variability in gene expression is explained by biological factors. The argument would be more convincing if the dropout rates and expression heterogeneity were minimal for highly expressed housekeeping genes. The authors are appropriately cautious in their interpretation and acknowledge the need for further validation.

(2) Heterogeneity across the board.

The authors focus on the relative heterogeneity in RNA abundance for surface proteins from the multicopy gene families vs core genes. While multicopy gene sequences do show significantly more cell-to-cell variability, there is still surprisingly high inequality of expression amongst genes in other classes including single copy housekeeping and ribosomal genes. Again the biological relevance of the comparison is uncertain and the authors acknowledge the need for further investigation.

This study provides some tantalising evidence that the expression of surface genes may vary substantially between individual parasites in a single clonal population. The study is also amongst the very first to apply scRNAseq to *T. cruzi*, so the broader data set will be an important resource for researchers in the field.

Comment on revised version:

The manuscript is significantly improved. The revised explanations and figures make several aspects of the data analysis and interpretation much clearer to me now. Thanks to the authors.

<https://doi.org/10.7554/eLife.105822.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

The authors aimed to assess the variability in the expression of surface protein multigene families between amastigote and trypomastigote Trypanosoma cruzi, as well as between individuals within each population. The analysis presented shows higher expression of multigene family transcripts in trypomastigotes compared to amastigotes and that there is variation in which copies are expressed between individual parasites. Notably, they find no clear subpopulations expressing previously characterised trans-sialidase groups. The mapping accuracy to these multicopy genes requires demonstration to confirm this, and the analysis could be extended further to probe the features of the top expressed genes and the other multigene families also identified as variable.

Strengths:

The authors successfully process methanol-fixed parasites with the 10x Genomics platform. This approach is valuable for other studies where using live parasites for these methods is logistically challenging.

Weaknesses:

The authors describe a single experiment, which lacks controls or complementation with other approaches and the investigation is limited to the trans-sialidase transcripts.

It would be more convincing to show either bioinformatically or by carrying out a controlled experiment, that the sequencing generated has been mapped accurately to different members of multigene families to distinguish their expression. If mapping to the multigene families is inaccurate, this will impact the transcript counts and downstream analysis.

We thank the reviewer for raising these important points.

We agree that the analysis of multigene families at the single-cell level is an important question, particularly given the heterogeneity observed across several of them. However, the aim of this short report is not to provide a comprehensive analysis of the entire experiment, but rather to focus on what we consider an important biological phenomenon observed in TcTS genes.

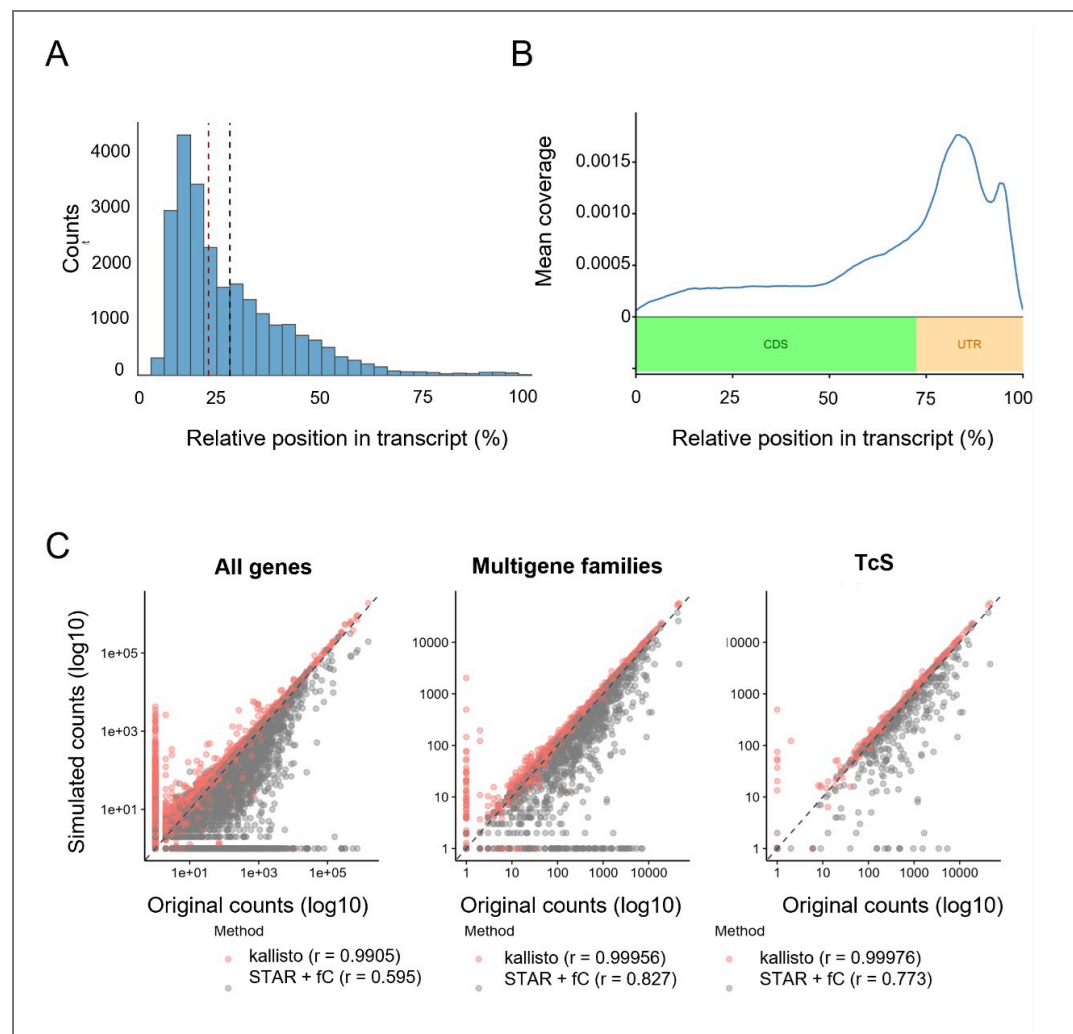
Regarding the mapping accuracy of the reads, we acknowledge that this can limit the disambiguation of highly similar multicopy transcripts. This is, in fact, a common challenge when analyzing transcriptomic data from *T. cruzi*.

To address this issue, we analyzed the sequence identity of the 3' ends of TcS transcripts (defined as the 3'UTR plus 20% of the CDS region). As shown in Author response image 1, these regions display a median sequence identity of approximately 25%, indicating that sufficient sequence divergence exists for mapping algorithms to use during read assignment.

In addition, it is important to note that kallisto, the software used in our analysis, was specifically designed to address multimapping reads through pseudoalignment combined with an expectation-maximization algorithm that probabilistically assigns reads across compatible transcripts.

To directly assess performance, we simulated reads from the *T. cruzi* transcriptome used in this study (3'UTRs plus 20% of the CDS regions) and compared two mapping/counting strategies: (a) transcriptome pseudoalignment using kallisto, and (b) genome alignment

followed by counting using STAR + featureCounts. The latter approximates the strategy implemented in Cell Ranger, the standard pipeline for quantifying expression levels from 10X Genomics single cell RNA-seq data. We found that kallisto recovered the simulated “true” counts with substantially higher accuracy than STAR + featureCounts (Pearson correlation: all genes, 0.991 vs 0.595; surface protein genes, 0.9996 vs 0.827; trans-sialidase (TcS) genes, 0.9998 vs 0.773). These results indicate that pseudoalignment is currently the optimal strategy for recovering the relative expression of highly similar gene family members (Author response image 1 C).



Author response image 1. (A) Distribution of pairwise sequence identity values calculated among the 3'-end regions of all transcripts (defined as the 3'UTR plus 20% of the coding sequence). **(B)** Distribution of read mapping coordinates over all multigene family transcripts normalized as percentage of the gene length **(C)** Scatter plots showing the correlation between estimated transcript counts obtained using kallisto (red) and STAR + featureCounts (grey) versus the corresponding simulated ground-truth values.

Reviewer #2 (Public review):

Summary:

This manuscript presents a valuable single-cell RNA-seq study on Trypanosoma cruzi, an important human parasite. It investigates the expression heterogeneity of surface proteins, particularly those from the trans-sialidase-like (TcS) superfamily, within amastigote and trypomastigote populations. The findings suggest a previously

underappreciated level of diversity in TcS expression, which could have implications for understanding parasite-host interactions and immune evasion strategies. The use of single-cell approaches to delve into population heterogeneity is strong. However, the study does have some limitations that need to be addressed.

*The focus on single-cell transcriptional heterogeneity in surface proteins, especially the TcS family, in *T. cruzi* is novel. Given the important role of these proteins in parasite biology and host interaction, the findings have potential significance.*

Strengths:

The key finding of heterogeneous TcS expression in trypomastigotes is well-supported. The analysis comparing multigene families, single-copy genes, and ribosomal proteins highlights the unusual nature of the variation in surface protein-coding genes.

Weaknesses:

While the manuscript identifies TcS heterogeneity, the functional implications of the different expression profiles remain speculative. The authors state it may reflect differences in infectivity, but no direct experimental evidence supports this.

The manuscript lacks any functional validation of the single-cell findings. For instance, do the trypomastigote subpopulations identified based on TcS expression exhibit differences in infectivity, host cell tropism, or immune evasion? Such experiments would greatly strengthen the study.

We thank the reviewer for their careful reading of the manuscript. We agree that obtaining experimental evidence on the influence of multiple multigene families would represent a significant advancement in the field. However, we would like to emphasize that this study is presented as a short communication centered on a specific and biologically relevant observation within a single multigene family. The aim of the manuscript is to highlight what we consider an important biological phenomenon that raises hypotheses to be tested in future work.

The influence of phenotypic heterogeneity and its possible advantages under environmental pressures has been previously proposed for *Trypanosoma cruzi*, related trypanosomatids, and other biological systems, ranging from bacteria to tumors (Seco-Hidalgo 2015, doi: 10.1098/rsob.150190 and Luzak 2021, doi: 10.1146/annurev-micro-040821-012953, for a comprehensive review on this topic). While the reviewer is correct in noting that our model does not demonstrate a functional role for TcTS heterogeneity, the experimental approaches required to address this question in a large multigene family are highly complex. This is particularly challenging in *T. cruzi*, where the study of multigene families is limited by the restricted set of available molecular biology tools (such as RNAi). Therefore, further experimental validation of these observations falls outside the scope of this short report.

In this revised version, we have included additional validation and clarification of the results, as well as a more explicit discussion of their limitations. In addition, we present a preliminary analysis exploring potential mechanisms that could coordinate the observed expression patterns of the TcTS family.

The authors identify a subpopulation of TcS genes that are highly expressed in many cells. However, it is unclear if these correspond to previously characterized TcS members with specific functions.

The TcS subgroup with a high frequency of detection comprises 31 genes, none of which belong to the catalytically active Group I trans-sialidases. Instead, this subgroup includes

members of Groups II, III, IV, V, VI, and VIII. This information has been added to Supplementary Table 3 and is now stated in the revised manuscript.

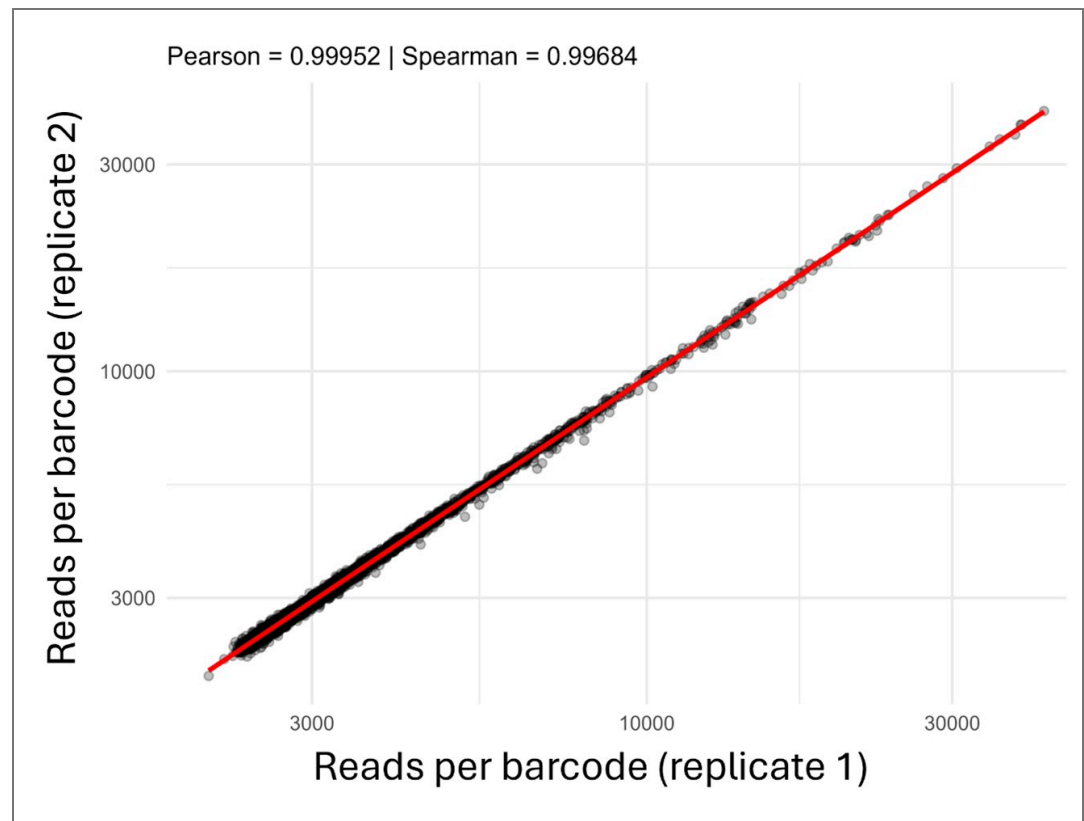
The authors hypothesize that observed heterogeneity may relate to chromatin regulation. However, the study does not directly address these mechanisms. There are interesting connections to be made with what they identify as the colocalization of genes within chromatin folding domains, but the authors do not fully explore this. It would be insightful to address these mechanisms in future work.

In response to the reviewer's and editorial team's request for additional mechanistic insight into the regulatory processes that may be involved in the observed patterns, we have expanded the revised manuscript to discuss how the genomic context of TcS loci could contribute to the observed heterogeneity in TcS expression. As noted in the original version of the manuscript, TcS genes and other surface-protein gene families are largely partitioned into discrete genomic compartments, whose expression has been reported to be regulated by epigenetic control of chromatin-folding domains (doi.org/10.1038/s41564-023-01483-y). However, we previously showed that TcS genes detected in a high proportion of cells are, in most cases, dispersed throughout the genome, arguing against a model in which their preferential expression results from colocalization within a small number of ubiquitously activated chromatin domains. In response to the reviewer's suggestion, we performed a more detailed analysis of the genomic locations of these TcS genes. We found that many of them are localized within the core compartment (new Figure 5). Because the core compartment is enriched for conserved, housekeeping genes that typically display more constitutive expression (doi.org/10.1038/s41564-023-01483-y), whereas the disruptive compartment is enriched for lineage-specific multigene families associated with variable, stage-specific, and recently reported stochastic expression (doi.org/10.1038/s41467-025-64900-2), our results are consistent with a model in which compartment-specific regulatory mechanisms (in addition to post-transcriptional regulation) influence the differential cellular expression of core-versus disruptive-located TcS genes. We have incorporated these results and discussion in the revised manuscript.

The merging of technical replicates needs further justification and explanation as they were not processed through separate experimental conditions. While barcodes were retained, it would be informative to know how well each technical replicate corresponds with the other. If both datasets were sequenced on the same lane, the inclusion of technical replicates adds noise to the analysis.

Regarding technical details, we now include the total number of mapped reads and average number of reads mapped per cell (new paragraph in the Methods section).

The technical replicates consist of a single Illumina library that was sequenced in two separate runs. As this approach is expected to be highly reproducible, we merged both runs into a single count table. To support this decision, we assessed the concordance between the two sequencing runs and observed an almost perfect correlation between them (Author response image 2).



Author response image 2. Correlation analysis of number of reads assigned to cells between technical replicate 1 and technical replicate 2.

While the number of cells sequenced (3192) seems reasonable, it's not clear how much the conclusions are affected by the depth of sequencing. A more detailed description of the sequencing depth and its impact on gene detection would be valuable.

We detected a mean of 1088 genes per cell. Based on the 15,319 annotated protein-coding genes in the reference genome, this represents 7.1% of the *T. cruzi* protein-coding gene complement detected in each cell.

Across the entire dataset, a total of 14,321 genes were detected in at least one cell, representing 93.5% of all annotated protein-coding genes. This suggests that our experiment captured a broad representation of the parasite's transcriptome.

This per-cell detection rate is characteristic of droplet-based scRNA-seq and is consistent with other trypanosomatid studies. For example, the *T. brucei* single-cell atlas (Hutchinson et al., 2021) reported a median detection of 1052 genes per cell. In the case of *T. cruzi*, the recently published pre-print of the *T. cruzi* single cell atlas from Laidlaw & García-Sánchez et al. reported a mean between 298 and 928 genes detected per cell (depending on the sample).

This information is now included in Methods.

While most of the methods are clear, the way in which the subsampled gene lists were generated could be more thoroughly described, as some details are not clear for the subsampling of single-copy genes.

The subsampling method was originally described in the Figure 2 legend; to better highlight this approach, we have now moved its description to the Methods section.

Some of the figures are difficult to interpret. For example, the color scaling in the heatmap of Supplementary Figure 3B is not self-explanatory and it is hard to extract meaningful conclusions from the graph.

We agree with the reviewer in this assessment. We have now modified the figures to be more self-explanatory and better reflect the conclusions.

Reviewer #3 (Public review):

*The study aimed to address a fundamental question in *T. cruzi* and Chagas disease biology - how much variation is there in gene expression between individual parasites? This is particularly important with respect to the surface protein-encoding genes, which are mainly from massive repetitive gene families with 100s to 1000s of variant sequences in the genome. There is very little direct evidence for how the expression of these genes is controlled. The authors conducted a single-cell RNAseq experiment of in vitro cultured parasites with a mixture of amastigotes and trypomastigotes. Most of the analysis focused on the heterogeneity of gene expression patterns amongst trypomastigotes. They show that heterogeneity was very high for all gene classes, but surface-protein encoding genes were the most variable. In the case of the trans-sialidase gene family, many sequence variants were only detected in a small minority of parasites. The biology of the parasite (e.g. extensive post-transcriptional regulation) and potential technical caveats (e.g. high dropout rates across the genome) make it difficult to infer what this might mean for actual protein expression on the parasite surface.*

We thank the reviewer for this important comment, highlighting a central challenge when studying trypanosomatid biology. We acknowledge that in most eukaryotes and particularly in *T. cruzi*, where there is a predominant role of post-transcriptional regulation, mRNA levels are not always directly correlated with protein abundance, as previously reported by us and others (10.1186/s12864-015-1563-8, 10.1128/msphere.00366-21, 10.1590/S0074-02762011000300002, 10.1042/bse0510031). Nevertheless, steady-state transcript levels obtained by RNA-seq remain informative for assessing differential gene expression, and this approach has been widely used as a proxy for the study of gene expression profiles in *T. cruzi* (10.7717/peerj.3017, 10.1371/journal.ppat.1005511, 10.1016/j.jbc.2023.104623, 10.3389/fcimb.2023.1138456, 10.1186/s13071-023-05775-4).

It's also interesting to note that recent proteomic analyses (10.1038/s41467-025-64900-2) have revealed substantial heterogeneity in the expression of surface proteins, including trans-sialidases, supporting the idea that the transcriptional heterogeneity we observe reflects a genuine biological feature that propagates to the protein level.

We have now added a sentence to the discussion acknowledging this limitation and discussed the results from Cruz-Saavedra, et al. in the revised manuscript.

(1) Limit of detection and gene dropouts

An average of ~1100 genes are detected per parasite which indicates a dropout rate of over 90%. It appears that RNA for the "average" single copy 'core' gene is only detected in around 3% of the parasites sampled (Figure 2c: ~100 / 3192). This may be comparable with some other trypanosome scRNAseq studies, but this still seems to be a major caveat to the interpretation that high cell-to-cell variability in gene expression is explained by biological rather than technical factors. The argument would be more convincing if the dropout rates and expression heterogeneity were minimal for well-known highly expressed genes e.g. tubulin, GAPDH, and ribosomal RNAs. Admittedly, in their Final Remarks, the authors are very cautious in their interpretation, but it would be good to see a more thorough discussion of technical factors that might explain the low detection rates and how these could be tested or overcome in future work.

(2) Heterogeneity across the board

The authors focus on the relative heterogeneity in RNA abundance for surface proteins from the multicopy gene families vs core genes. While multicopy gene sequences do show more cell-to-cell variability, the differences (Figure 2D) are roughly average Gini values of 0.99 vs 0.97 (single copy) or 0.95 (ribosomal). Other studies that have applied similar approaches in other systems describe Gini values of < 0.2-0.25 for evenly expressed "housekeeping" genes (PMIDs 29428416, 31784565). Values observed here of >0.9 indicate that the distribution for all gene classes is extremely skewed and so the biological relevance of the comparison is uncertain.

We recognize the limitations imposed by gene dropout in our data, as highlighted by the reviewer. Unfortunately, gene dropout is an inherent limitation of 10x genomics data. Trypanosomatids are not an exception in this regard, and the general metrics of the single-cell RNA-seq data in other reports are equivalent to those obtained in our experiment.

Despite this important limitation, we believe that our comparative analyses (the contrast between TcS and ribosomal protein expression) provide valuable insights into a biological phenomenon with potential functional relevance for the parasite. Furthermore, we are actively working on generating single-cell RNA-seq data using alternative methodologies that improve gene dropout rates. We anticipate that these future studies will help clarify the extent of the phenomenon described in this work.

Our results reveal a small subset of TcS genes that are frequently detected across cells, a pattern that is not compatible with random detection unless these genes were highly expressed and preferentially captured by random sampling. However, as shown in Figure 4b, many genes expressed at comparable levels are not detected at high frequencies. In line with this, Figure 4c shows that within individual cells, the detected TcS genes exhibit similar expression levels. Finally, we confirmed that this frequently detected subset shows high read counts at the bulk RNA-seq level (Figure 4 - Figure Supplement 1), consistent with the fact that these TcS are frequent in the population even when they are not specially highly expressed within each cell. Taken together, these findings argue against a purely random sampling of TcS genes and support the interpretation that this pattern reflects an underlying biological feature. We agree that further validation will be required. Accordingly, since the initial submission, we have been careful to frame our conclusions conservatively, explicitly noting that dropout remains a limitation of these data that could influence the observed patterns. In the revised version, we have strengthened this point by including a specific statement in the final remarks. Our interpretation is presented as a working hypothesis that is fully compatible with the observations reported here and may be informative for the field. To better reflect this reasoning, we have revised Figure 4b, expanded the discussion, and explicitly included this limitation in the final remarks of the revised manuscript.

*Nevertheless, this study does provide some tantalising evidence that the expression of surface genes may vary substantially between individual parasites in a single clonal population. The study is also amongst the very first to apply scRNAseq to *T. cruzi*, so the broader data set will be an important resource for researchers in the field.*

We thank the reviewer for highlighting the relevance of our study and for their positive assessment of the potential significance of these observations. We also agree that the dataset generated here may represent a useful resource for the community.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) In Figures 1c and 1d, it would be useful to include the genes as the plot titles.

We agree with the reviewer that including gene names in the plot makes the panels more self-explanatory. We have added gene names to the updated version of Figure 1.

(2) Can you include the read lengths of the sequencing and whether this is sufficient to map accurately to very similar genes of the same multigene family? As stated in the public summary, this would make the data far more convincing as standard 10x chromium cannot distinguish similar gene copies unless a longer read 2 is used. Given that only the 3' end is targeted, is this enough to distinguish the TcS and other multigene family transcripts?

We thank the reviewer for raising this important point. We agree that short 3' biased reads can limit the disambiguation of highly similar multicopy transcripts. This is, in fact, a common challenge when analyzing transcriptomic data from *T. cruzi*.

To address this issue, we analyzed the sequence identity of the 3' ends of TcS transcripts (defined as the 3'UTR plus 20% of the CDS region). As shown in Author response image 1, these regions display a median sequence identity of approximately 25%, indicating that sufficient sequence divergence exists for mapping algorithms to use during read assignment.

In addition, it is important to note that kallisto, the software used in our analysis, was specifically designed to address multimapping reads through pseudoalignment combined with an expectation-maximization algorithm that probabilistically assigns reads across compatible transcripts.

To directly assess performance, we simulated reads from the *T. cruzi* transcriptome used in this study (3'UTRs plus 20% of the CDS regions) and compared two mapping/counting strategies: (a) transcriptome pseudoalignment using kallisto, and (b) genome alignment followed by counting using STAR + featureCounts. The latter approximates the strategy implemented in Cell Ranger, the standard pipeline for quantifying expression levels from 10X Genomics single cell RNA-seq data. We found that kallisto recovered the simulated “true” counts with substantially higher accuracy than STAR + featureCounts (Pearson correlation: all genes, 0.991 vs 0.595; surface protein genes, 0.9996 vs 0.827; trans-sialidase (TcS) genes, 0.9998 vs 0.773). These results indicate that pseudoalignment is currently the optimal strategy for recovering the relative expression of highly similar gene family members (Author response image 1C).

The length of the R2 read (91bp) was included in Methods (line 411).

(3) It is stated that 'single copy' genes also include 'low copy number genes'. What does this include exactly? Is it more accurate to say non-surface protein genes?

The distinction we aim to make is between multigene families and the rest of the genome. Most multigene families encode surface proteins, but not all surface protein genes belong to multigene families. To clarify this point we included a sentence in methods to reflect that when we describe “surface proteins” we are referring to surface proteins coded by multigene families (line 453). In addition, long-read genomic DNA sequencing and assembly have revealed that many genes previously believed to be single-copy are actually duplicated at low copy numbers (doi.org/10.1099/mgen.0.000177). For this reason, we extend the concept of “single-copy” genes to include those that have only a few duplicates.

(4) It is stated in line 127 that TcS have particular high heterogeneity - it does not look that way by eye compared to the other multigene families. Can statistic be used to prove this, or simply state the decision was made to focus on the TcS?

As noticed by the reviewer, all multigene families show significantly higher heterogeneity compared to single-copy genes, as stated in the text and shown in figure legends from Figure

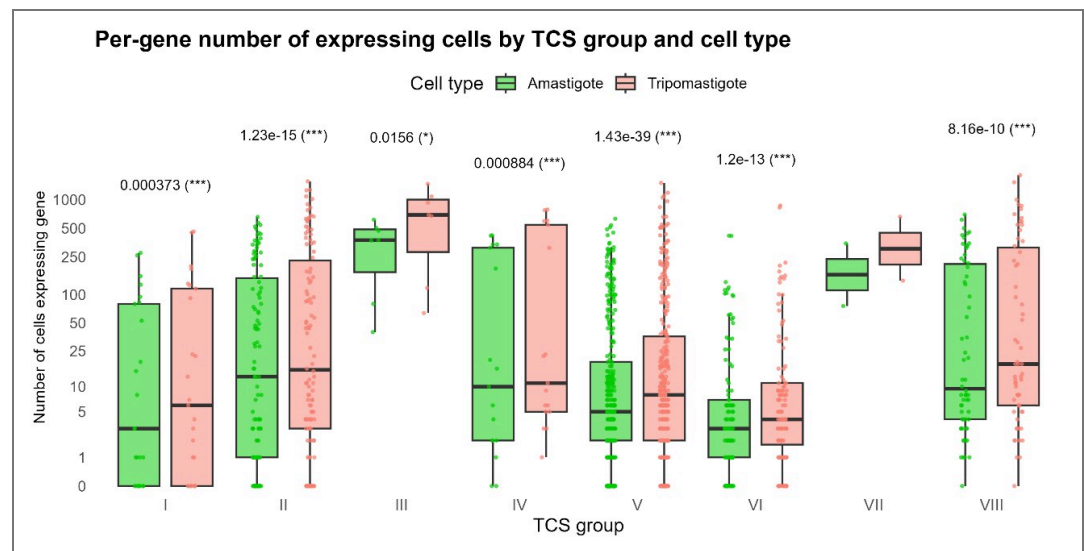
2, Supplementary Figure 1 and the new Supplementary Table 2.

That said, it was not the statistical results that guided our decision to focus on TcS, but rather their well-established biological relevance in *T. cruzi*. As suggested, we have now emphasized this rationale more clearly in the revised text (lines 160-167).

Besides, recent work has shown that TcS genes exhibit a bimodal distribution of expression levels using bulk RNA-seq data, in contrast to core genes and other multigene families (doi.org/10.1038/s41467-025-64900-2, doi.org/10.1038/s41564-023-01483-y). This distinct regulatory behavior further justifies our decision to examine TcS separately.

(5) Expression of different TcS has been investigated between the different life cycle stages for a few individual genes previously (Freitas et al). Can the authors not extend this investigation to all the genes detect by scRNA-seq here to demonstrate those with higher/lower expression in amastigotes vs trypomastigotes building on Figure 2A? Are particular groups linked to either stage?

We performed this analysis and did not observe any correlation between TcS groups and life cycle stage. In all cases TcS were more frequently detected in trypomastigotes. This difference was statistically significant for all groups except group VII, likely due to the low number of genes analyzed in this group (Author response image 3).



Author response image 3. Per-gene number of expressing cells by TcS group and life-stage. Boxplots show, for each TcS group (I–VIII), the distribution across genes of the number of cells in which the gene is detected. Each point represents a single TcS; Amastigote cells: green points/boxes, Trypomastigote cells: salmon points/boxes. The y-axis is on log10 scale. Asterisks indicate statistically significant differences from the comparison between Amastigote and Trypomastigote within each TcS group, assessed using a paired two-sided Wilcoxon signed-rank test: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

(6) What exactly is the Z-score shown in Figure 2B?

In this analysis num_multigene represents the number of multigene family genes detected in each individual cell. For every cell, we counted how many genes from our predefined multigene family gene list has detectable expression (more than zero UMI counts); in the UMAP plot, this value is reflected by the size of each point. On the other hand, z_multigene captures the relative expression level of multigene family genes within each cell. This metric is calculated by summing the UMI counts of all multigene family genes per cell and then standardizing this value across the dataset using a z-score transformation, such that positive

values reflect above-average multigene family expression and negative values reflect below-average levels. In the UMAP plot, this metric determines the color scale of each point. Taking together num_multigene and z_multigene allow us to distinguish cells that express multigene family genes broadly (high gene counts), strongly (high relative expression), both, or neither, and to relate these patterns to identified cell populations.

We included a short description in legend of the new version of Figure 2 (lines 176-180).

(7) For the reclustering of trypomastigotes based on TcS genes alone, please show the UMAP and discuss why the resolution giving two clusters is chosen? I assume increasing the resolution does not reveal clusters of cells express one of the 8 groups of TcS for example?

We appreciate the reviewer's suggestion. In this analysis, our goal was to test whether the phenotypic heterogeneity previously reported in trypomastigotes could be recapitulated using TcS genes alone, as prior studies described two major transcriptomic phenotypes within this stage.

Increasing the clustering resolution did not reveal subclusters corresponding to the eight TcS sequence groups. This might reflect the fact that these groups are defined based on sequence similarity rather than on expression patterns, as noted by Freitas et al. (doi:10.1371/journal.pone.0025914).

(8) In Figure 4B, there may be an upward trend in the level of expression and the number of cells a transcript is detected in? It would be worth showing this is or is not the case with statistics if possible.

The number of genes detected in a high proportion of cells is low, which limits the statistical power of this analysis. Also, substantial dispersion is observed within the 0-5% interval. Nevertheless, this figure is presented primarily to highlight that a considerable number of highly expressed genes are detected in only a small fraction of cells. If expression level were the main determinant of detection frequency across cells, one would expect very few highly expressed genes to fall within the 0-5% interval. Contrary to this expectation, among the 50 highest expressed TcS genes, 62% are detected in fewer than 5% of cells, and even among the top 10 most highly expressed TcS genes, 40% fall within this lowest detection group. To facilitate this interpretation, we modified the figure (new Figure 4b) to explicitly highlight the top 50 most expressed TcS genes and incorporated this discussion into the main text of the revised manuscript (lines 244-251), making the conclusion clearer to the reader.

(9) Do the cells group instead by expression of any of the other multigene families not investigated in detail?

It is possible that additional transcriptional substructure among trypomastigotes is driven by the expression of other multigene families beyond TcS. In this short report (with limited number of figures, words, etc.), we focused specifically on the trans-sialidase family as discussed earlier. A more comprehensive analysis including other large surface gene families (MASPs, mucins, GP63) is planned as part of ongoing work and will be presented in future reports.

Reviewer #2 (Recommendations for the authors):

This reviewer suggests the conduction of functional experiments in follow-up studies to establish links between TcS expression profiles and parasite behavior and into potential regulatory mechanisms responsible for the observed TcS heterogeneity, particularly focusing on epigenetic modifications. It would be interesting to correlate the highly expressed TcS members identified here with previously characterized TcS isoforms and provide more description regarding which particular groups and TcS members are

driving the findings. It would benefit from further clarification regarding sequencing depth, technical replication merging, subsampling, and specific parameters for alignment methods and more information regarding the specific statistical tests and their applicability to the data.

This is a promising single-cell study with potentially high significance. The manuscript is well-written, and the analyses are reasonably well-executed. However, the current manuscript is limited by a lack of functional validation and mechanistic insights. The addition of further analyses and experiments, as suggested, will strengthen the conclusions and increase the impact of the work.

We thank the reviewer for their careful reading of the manuscript. As suggested, we have performed additional validation and clarification of the results, as well as a more explicit discussion of their limitations. In addition, we have included a preliminary analysis exploring potential mechanisms that could be coordinating the observed expression patterns of the TcS family (see below). Even though we consider relevant and interesting to experimentally validate these results, given the inherent difficulties in studying multigene families in *T. cruzi*, an organism with a very limited set of molecular biology tools (such as RNAi), further experimental validation of these observations is outside of the scope of this short report.

Regarding the reviewer's question, we studied if any TcS subgroup could be driving our observations. However, we did not find any correlations indicating that a particular group was associated with any of our findings. We now include TcS group information to Supplementary Table 3.

Regarding technical details, we now included the total number of mapped reads (line 422) and average number of reads mapped per cell (new paragraph in the Methods section, line 432-436).

The technical replicates consist of a single Illumina library that was sequenced in two separate runs. As this approach is expected to be highly reproducible, we merged both runs into a single count table, as stated in line 424. To support this decision, we assessed the concordance between the two sequencing runs and observed an almost perfect correlation between them (Author response image 2).

The subsampling method was originally described in the Figure 2 legend; to better highlight this approach, we have now moved its description to the Methods section (line 456).

The specific kallisto parameters used are stated in Methods (line 418-419). We now included that default options were used unless otherwise specified (line 419-420).

In response to the reviewer's and editorial team's request for additional mechanistic insight into the regulatory processes that may be involved in the observed patterns, we have expanded the revised manuscript to discuss how the genomic context of TcS loci could contribute to the observed heterogeneity in TcS expression. As noted in the original version of the manuscript, TcS genes and other surface-protein gene families are largely partitioned into discrete genomic compartments, whose expression has been reported to be regulated by epigenetic control of chromatin-folding domains (doi.org/10.1038/s41564-023-01483-y). However, we previously showed that TcS genes detected in a high proportion of cells are, in most cases, dispersed throughout the genome, arguing against a model in which their preferential expression results from colocalization within a small number of ubiquitously activated chromatin domains. In response to the reviewer's suggestion, we performed a more detailed analysis of the genomic locations of these TcS genes. We found that many of them are localized within the core compartment (new Figure 5). Because the core compartment is enriched for conserved, housekeeping genes that typically display more constitutive expression (doi.org/10.1038/s41564-023-01483-y), whereas the disruptive compartment is

enriched for lineage-specific multigene families associated with variable, stage-specific, and recently reported stochastic expression (doi.org/10.1038/s41467-025-64900-2), our results are consistent with a model in which compartment-specific regulatory mechanisms (in addition to post-transcriptional regulation) influence the differential cellular expression of core-versus disruptive-located TcS genes. We have incorporated these results and discussion in line 301-313 of the revised manuscript.

Reviewer #3 (Recommendations for the authors):

(1) The authors consistently refer to gene "expression" but somewhere they should acknowledge that in trypanosomes RNA abundance is less predictive of protein than in most other organisms.

We thank the reviewer for this important comment, highlighting a central challenge when studying trypanosomatid biology. We acknowledge that in most eukaryotes and particularly in *T. cruzi*, where there is a predominant role of post-transcriptional regulation, mRNA levels are not always directly correlated with protein abundance, as previously reported by us and others (10.1186/s12864-015-1563-8, 10.1128/msphere.00366-21, 10.1590/S0074-02762011000300002, 10.1042/bse0510031). Nevertheless, steady-state transcript levels obtained by RNA-seq remain informative for assessing differential gene expression, and this approach has been widely used as a proxy for the study of gene expression profiles in *T. cruzi* (10.7717/peerj.3017, 10.1371/journal.ppat.1005511, 10.1016/j.jbc.2023.104623, 10.3389/fcimb.2023.1138456, 10.1186/s13071-023-05775-4).

It's also interesting to note that recent proteomic analyses (10.1038/s41467-025-64900-2) have revealed substantial heterogeneity in the expression of surface proteins, including trans-sialidases, supporting the idea that the transcriptional heterogeneity we observe reflects a genuine biological feature that propagates to the protein level.

We have now added a sentence to the discussion acknowledging this limitation and discussed the results from Cruz-Saavedra, et al. in line 266-271 of the revised manuscript.

(2) Line 29, in the abstract there is a strong statement that T. cruzi "does not employ antigenic variation". I don't think there is much evidence either way if we are thinking about antigenic variation in the broad sense rather than the extreme model of T. brucei VSG switching. Later in the abstract they state that "no recurrent combinations of TcS genes were observed between individual cells in the population", which sounds very much like a form of antigenic variation.

We agree with the reviewer. Indeed, we meant to state that *T. cruzi* does not employ an antigenic variation mechanism such as the one from *T. brucei*. We change this statement as suggested in lines 28 - 32.

(3) Line 29, "relies on a diverse array of cell-surface-associated proteins encoded by large multi-copy gene families (multigene families) essential for infectivity and immune evasion" and lines 55-58 "T. cruzi infection relies on a heterogeneous set of membrane proteins, encoded mainly by large multigene families ... most of which are involved in infection, tropism, and immune evasion". It would be worth adding a bit more detail on the nature and strength of the evidence that Tc "relies on" these various genes or that they are "essential" for infectivity, tropism, and immune evasion.

Because the journal's short format imposes word limits, we strengthened the original statement by adding specific references that document genomic, transcriptomic and functional evidence linking the major multigene families to infectivity, tropism and immune evasion (doi.org/10.1371/journal.pone.0025914; doi.org/10.1038/nrmicro1351; doi.org/10.1128/iai.05329-11; doi.org/10.1093/nar/gkp172, doi.org/10.1371/journal.ppat.1006767), in line 77.

(4) Line 89, 1088 genes detected per cell - what is this as a % of genes in the genome?

We detected a mean of 1088 genes per cell. Based on the 15,319 annotated protein-coding genes in the reference genome, this represents 7.1% of the *T. cruzi* protein-coding gene complement detected in each cell.

Across the entire dataset, a total of 14,321 genes were detected in at least one cell, representing 93.5% of all annotated protein-coding genes. This suggests that our experiment captured a broad representation of the parasite's transcriptome.

This per-cell detection rate is characteristic of droplet-based scRNA-seq and is consistent with other trypanosomatid studies. For example, the *T. brucei* single-cell atlas (Hutchinson et al., 2021) reported a median detection of 1052 genes per cell. In the case of *T. cruzi*, the recently published pre-print of the *T. cruzi* single cell atlas from Laidlaw & García-Sánchez et al. reported a mean between 298 and 928 genes detected per cell (depending on the sample).

This information is now included in Methods (line 435).

(5) Line 93-94, how many cells were assigned to clusters 0 and 1?

Cluster 0 had 2201 cells and cluster 1 had 824 cells assigned. We have now included these specific numbers in new version of the manuscript (line 114).

(6) Line 96, cluster 2 ama-trypo transitioning parasites - were these observable by microscopy?

We did not perform microscopy specifically to observe or quantify the putative ama/trypo transitioning subpopulation: microscopy was only used as a pre-experiment quality check to verify cell morphology and viability. The inference that cluster 2 reflects ama/trypo transitioning parasites is drawn from the transcriptomic profile (particularly from the pattern of stage-associated marker expression observed in that cluster) and should be considered a hypothesis generated by the data, that merits further analysis, as stated in the manuscript.

(7) Line 106-107, "As expected, single-copy gene expression is high in both amastigotes and trypomastigotes and similar on average between both cell types".

(8) Why as expected? For a broad journal it would be useful to explain this. Amastigotes are replicative and trypomastigotes are not, so would we not expect to see some differences that reflect this?

(9) What do you mean by the expression being "high"? High compared to what?

(10) "Similar on average between both cell types". This does not seem concordant with Figure 1a showing a highly significant difference between ama and trypo.

We thank the reviewer for this helpful request for clarification for broader readers and the observations regarding global expression of single copy and multigene family genes.

Figure 2a is intended as an experimental control where we show that our 10X Genomics data shows the previously reported upregulation of surface protein genes in trypomastigotes. We have now modified the text in order to highlight this (line 129). In turn, Supplementary Figure 1a is shown as a control that this upregulation is not a general feature of trypomastigote cells.

Regarding comment 9, what we meant is that single-copy genes display relatively high expression in both amastigotes and trypomastigotes compared with surface protein-coding genes (see expression values in Figures 2a and Supplementary Figure 1a).

Finally, differential expression between amastigotes and trypomastigotes at the transcriptomic level has been previously studied and has shown that most single copy genes do not show variation, explaining the overall pattern of Supplementary Figure 1a where average expression is similar between stages (mean fold change = 1.1). This is likely due to the fact that these genes are related to basic cellular functions. Genes related to stage specific functions such as replication in amastigotes or normalization effects may be causing the slight, but statistically significant increase observed in overall expression in amastigotes. This contrasts with the pattern observed for multigene families where there is a clear overexpression in trypomastigotes (mean fold change = 1.5).

As observations commented on questions 9 and 10 have been described in previous studies and are not novel nor key points in our results, we decided not to focus on them and modified the text accordingly in lines 129-135.

(11) Line 110, "with high variation". What does "high variation" mean here? Compared to what? For the two metrics (n cells +ve for each gene and total expression level) can they give an average and the SD? It would be useful to know how many parasites the "average" surface (and core) gene is expressed in, or more precisely for which the RNA is above the limit of detection.

We refer to the comparison with the expression profile observed for single-copy genes. This point has now been clarified in the text, and we have included the mean and standard deviation for both TcS multigene family genes and single-copy genes in trypomastigotes for both metrics in the Figure 2 legend. The average and distribution of the number of cells in which each gene is detected are shown in Figure 2c and Supplementary Figure 1a. We also added a reference to this panel at the point in the text where the phenomenon is first described.

(12) Line 134, Figure 2b legend needs more detail - what are num_multigene and z_multigene?

Please see our response to Reviewer 1, Question 6. We have now added a clarification to the legends of Figure 1 and Supplementary Figure 1.

(13) Figure 2c, correct the y-axis legend because it implies your values are log10 transformed. Also, it would be useful to have more markers on the y axis so the reader can better estimate the data ranges.

We thank the reviewer for this observation. We have now corrected the y-axis label and markers.

(14) If the y-axis of Figure 2D started at 0 instead of 0.8 and if Lorenz curves were provided then the reader would probably get a fuller sense of the expression heterogeneity in the dataset. The legend states the differences are statistically significant but the actual p-values are not shown.

(15) Line 142-3, more precision is needed on the p-values.

We thank the reviewer for this helpful suggestion. We agree that Lorenz curves provide a clearer representation of expression heterogeneity than the previous plot. Accordingly, we have replaced the original panel (Figure 2d) with Lorenz curves for the groups under comparison, and have made the same change in Supplementary Figure 1d. In addition, we have included gini index values and p-values for all comparisons in Supplementary Table 2.

(16) Figure 3, as in Figure 1a it would be useful to add another UMAP plot to show the two trypan subpopulations.

We thank the reviewer for this suggestion. We have now updated Figure 3 to include a UMAP plot showing the two trypomastigote subpopulations.

(17) What is the observed proportion of broad vs slender trypomastigote morphologies for Dm28c? To be consistent with the speculation at line 162 then wouldn't it need to be approximately 50-50?

The proportions of each trypomastigote subpopulation in the DM28c strain are currently unknown. The only available relevant data come from Brener, 1965 (doi.org/10.1080/00034983.1965.11686277), in which this strain was not included. In the strains analyzed in that study, the relative proportions of broad and slender trypomastigote morphologies were highly variable: across seven strains, broad forms ranged from 18.0% to 77.3%, while slender forms ranged from 2.3% to 71.6%. Given this wide variability and the lack of DM28c-specific data, we cannot assume any expected proportion for this strain.

(18) Line 170, please state how many genes are in the TcS subgroup mentioned here. This is an interesting finding - does this include mostly catalytically active trans-sialidase genes or is it a mixture from across all the subfamilies?

The TcS subgroup with a high frequency of detection comprises 31 genes, none of which belong to the catalytically active Group I trans-sialidases. Instead, this subgroup includes members of Groups II, III, IV, V, VI, and VIII. This information has been added to Supplementary Table 3 and is now stated in the revised manuscript (lines 227 - 228).

(19) Line 175-176, "Gene dropouts might favor random patterns of gene family's detection in scRNA-seq experiments, particularly affecting genes with low expression" - I'm not sure if the authors mean the detection of a gene (or not) in an individual parasite is truly random (pure luck) or whether the term stochastic would be more appropriate because they seem to be referring to randomness around a certain threshold of RNA abundance/stability? They go on to rule this out, at least for TcS genes, essentially arguing that they have something resembling an ON or OFF pattern rather than a spectrum of expression levels. This is potentially very important and could advance the field in a major way, but the fact that so many core and ribosomal genes, which 'should' be always ON, cannot be detected in most cells is a concern. A version of Figure 4B for core and ribosomal genes could be informative - do they show a different pattern to TcS?

Our results reveal a small subset of TcS genes that are frequently detected across cells, a pattern that is not compatible with random detection unless these genes were highly expressed and preferentially captured by random sampling. However, as shown in Figure 4b, many genes expressed at comparable levels are not detected at high frequencies. In line with this, Figure 4c shows that within individual cells, the detected TcS genes exhibit similar expression levels. Finally, we confirmed that this frequently detected subset shows high read counts at the bulk RNA-seq level (Supplementary Figure 2), consistent with the fact that these TcS are frequent in the population even when they are not specially highly expressed within each cell. Taken together, these findings argue against a purely random sampling of TcS genes and support the interpretation that this pattern reflects an underlying biological feature. We agree that further validation will be required. Accordingly, since the initial submission, we have been careful to frame our conclusions conservatively, explicitly noting that dropout remains a limitation of these data that could influence the observed patterns. In the revised version, we have strengthened this point by including a specific statement in the final remarks. Our interpretation is presented as a working hypothesis that is fully compatible with the observations reported here and may be informative for the field. To better reflect this reasoning, we have revised Figure 4b, expanded the discussion, and explicitly included this limitation in the final remarks of the revised manuscript.

(20) Line 238-9, Add details of removing extracellular epimastigotes after cell infections.

Only cellular trypomastigotes collected from the supernatant on day 6 were used for the secondary infection, at a 10:1 parasite-to-cell ratio. After 24 hours, the cultures were washed twice with PBS to remove any remaining extracellular parasites. Under these conditions, i.e. using exclusively trypomastigotes, at this infection ratio, and maintaining the cultures in mammalian medium, we do not expect the presence or survival of extracellular epimastigotes. We have included a sentence in the Methods section clarifying this information in the revised version of the manuscript, line 382.

(21) Line 260, was methanol used to directly resuspend the parasite pellet, or was it resuspended first e.g. in a small volume of PBS?

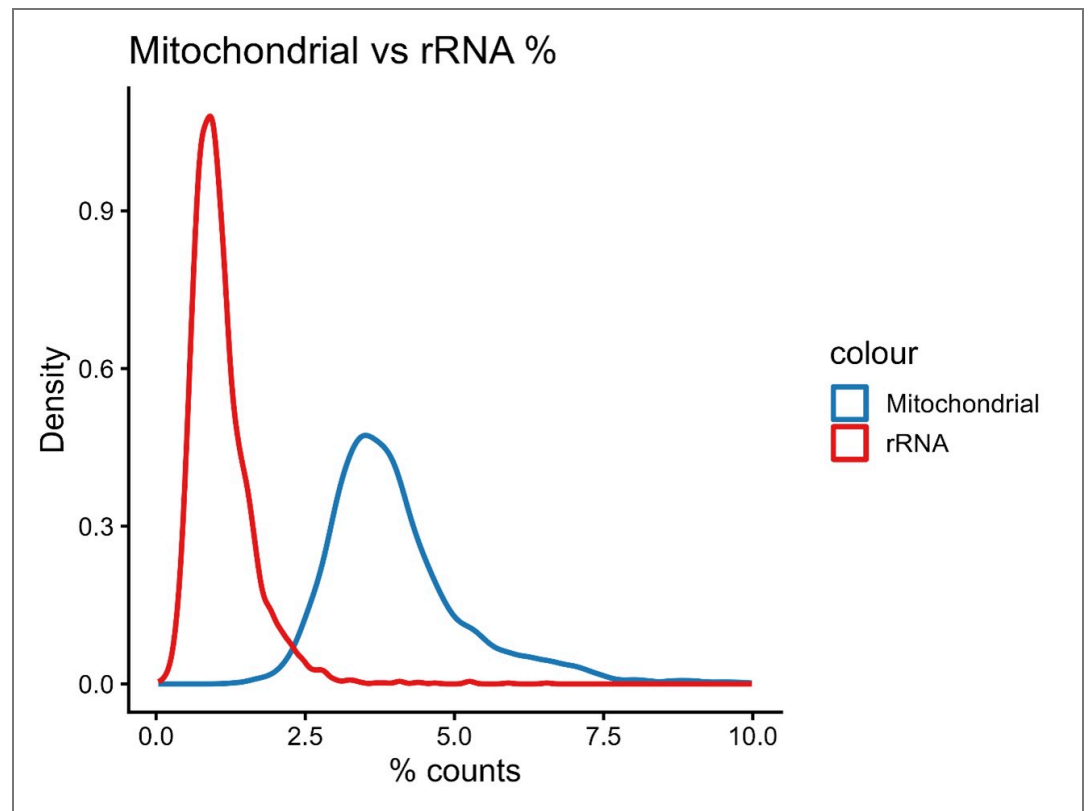
As described in lines 250-257 of the original manuscript, parasites were washed and resuspended in DPBS before methanol fixation. Methanol fixation was then carried out according to the 10X Genomics Methanol Fixation Protocol. We have now emphasized this more clearly in the revised text in line 400.

(22) What was the doublet rate?

We identified and removed 41 doublets, all belonging to cluster 2, and retained 3,151 singlets for downstream analysis (total cells before removal = 3,192). The resulting doublet rate was 1.28%. We have included a sentence in the Methods section clarifying this information in the revised version of the manuscript, line 439 -440.

(23) What was the frequency of rRNA and kDNA-derived reads?

Approximately 4.02% of the reads were derived from kDNA sequences, while 1.10% corresponded to rRNA-derived reads (Author response image 4).



Author response image 4. Percentage of mitochondrial and ribosomal rRNA derived reads.

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